Project File

Web Programming using

Python and JavaScript

Name: Aayush

Course: B-Tech CSE Core

Year: 2nd Year, 4th Semester

Roll No: 2401010042

Submitted To:

Ms. Mansi Kajal

EXPERIMENT – I

❖ **Aim :**

Creating a **Personal Portfolio Web Application** using **HTML, CSS, and JavaScript**.

❖ **Tasks :**

- **Project Setup & Introduction** – Create project folder and basic linked files.
- **HTML Structure** – Build portfolio sections using semantic HTML.
- **Styling with CSS** – Apply layout, colors, spacing, and neat design.
- **Projects & Skills Display** – Show skills and projects clearly on the page.
- **JavaScript Interactivity** – Add basic interactive functionality using JavaScript.
- **Contact Form (Bonus)** – Create a form with user submission feedback.

❖ **Tech Stack Used :**

- **HTML** , **CSS** and **JavaScript**

❖ Code :

➤ **HTML CODE :**

```
<!--  -->
<!-- Name: Aayush
Project: Personal Portfolio -->
<!--  -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Aayush Portfolio</title>
  <link rel="stylesheet" href="style.css">
```

```
</head>

<body>

<header>
  <h1>Aayush</h1>
  <p>I am learning Web Development</p>
</header>

<nav>
  <a href="#skills">Skills</a>
  <a href="#projects">Projects</a>
  <a href="#contact">Contact</a>
</nav>

<section class="welcome">
  <h2>Welcome</h2>
  <p>This is my beginner portfolio website.</p>
  <button onclick="showMessage()">Click Me</button>
</section>

<section id="skills">
  <h2>
    My Skills
    <button onclick="toggleSection('skillsContent')">Hide / Show</button>
  </h2>
  <ul id="skillsContent">
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript (Beginner)</li>
  </ul>
```

```
</section>

<section id="projects">
  <h2>
    My Projects
    <button onclick="toggleSection('projectsContent')">Hide / Show</button>
  </h2>

  <div id="projectsContent">
    <div class="card">
      <h3>Portfolio Website</h3>
      <p>This is my personal portfolio made using HTML, CSS and JavaScript.</p>
    </div>

    <div class="card">
      <h3>Simple HTML Page</h3>
      <p>A basic web page created using only HTML.</p>
    </div>
  </div>
</section>

<section id="contact">
  <h2>Contact Me</h2>

  <form onsubmit="submitForm(); return false;">
    <label>Name:</label>
    <input type="text" required>

    <label>Email:</label>
    <input type="email" required>
```

```
        <label>Message:</label>
        <textarea rows="4" required></textarea>

        <button type="submit">Submit</button>
    </form>

    <p id="result"></p>

    <a href="https://github.com/83Aayush" target="_blank">GitHub</a> |
    <a href="https://www.linkedin.com/in/aayush-83/" target="_blank">LinkedIn</a>
</section>

<footer>
    <p>© 2026 Aayush</p>
</footer>

<button id="scrollBtn" onclick="scrollToTop()">⬆ </button>

<script src="script.js"></script>
</body>
</html>
```

➤ **CSS CODE :**

```
@import url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap');
* {
    box-sizing: border-box;
}
body {
    margin: 0;
    font-family: 'Poppins', sans-serif;
    background: linear-gradient(to right, #fbc2eb, #a6c1ee);
```

```
}
header {
   background: purple ;
   color: white;
   padding: 40px 20px;
   text-align: center;
}
header h1 {
   margin: 0;
   font-size: 38px;
}
header p {
   font-size: 18px;
   opacity: 0.9;
}
nav {
   background-color: #ffffff;
   padding: 12px;
   text-align: center;
   box-shadow: 0 4px 10px rgba(0,0,0,0.1);
}
nav a {
   color: #0a89dd;
   text-decoration: none;
   margin: 0 15px;
   font-weight: 600;
}
nav a:hover {
   color: blueviolet;
}
```

```
section {
    background-color: white;
    max-width: 900px;
    margin: 30px auto;
    padding: 25px;
    border-radius: 12px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.15);
}
.welcome {
    text-align: center;
}
.welcome h2 {
    font-size: 28px;
}
button {
    background: linear-gradient(to right, #6a11cb, #2575fc);
    color: white;
    border: none;
    padding: 10px 18px;
    border-radius: 25px;
    cursor: pointer;
    font-size: 14px;
    transition: 0.3s;
}
button:hover {
    transform: scale(1.05);
}
.card {
    background: #f9f9f9;
    border-radius: 10px;
    padding: 20px;
```

```
    margin: 15px 0;
    transition: 0.3s;
}
.card:hover {
    transform: translateY(-5px);
    box-shadow: 0 8px 20px rgba(0,0,0,0.15);
}
ul {
    list-style: none;
    padding: 0;
}
ul li {
    background: #eef3ff;
    margin: 8px 0;
    padding: 10px;
    border-radius: 6px;
    font-weight: 500;
}
form {
    margin-top: 15px;
}
input, textarea {
    width: 100%;
    padding: 10px;
    margin: 8px 0;
    border-radius: 8px;
    border: 1px solid #ccc;
    font-family: 'Poppins', sans-serif;
}
input:focus, textarea:focus {
    outline: none;
```

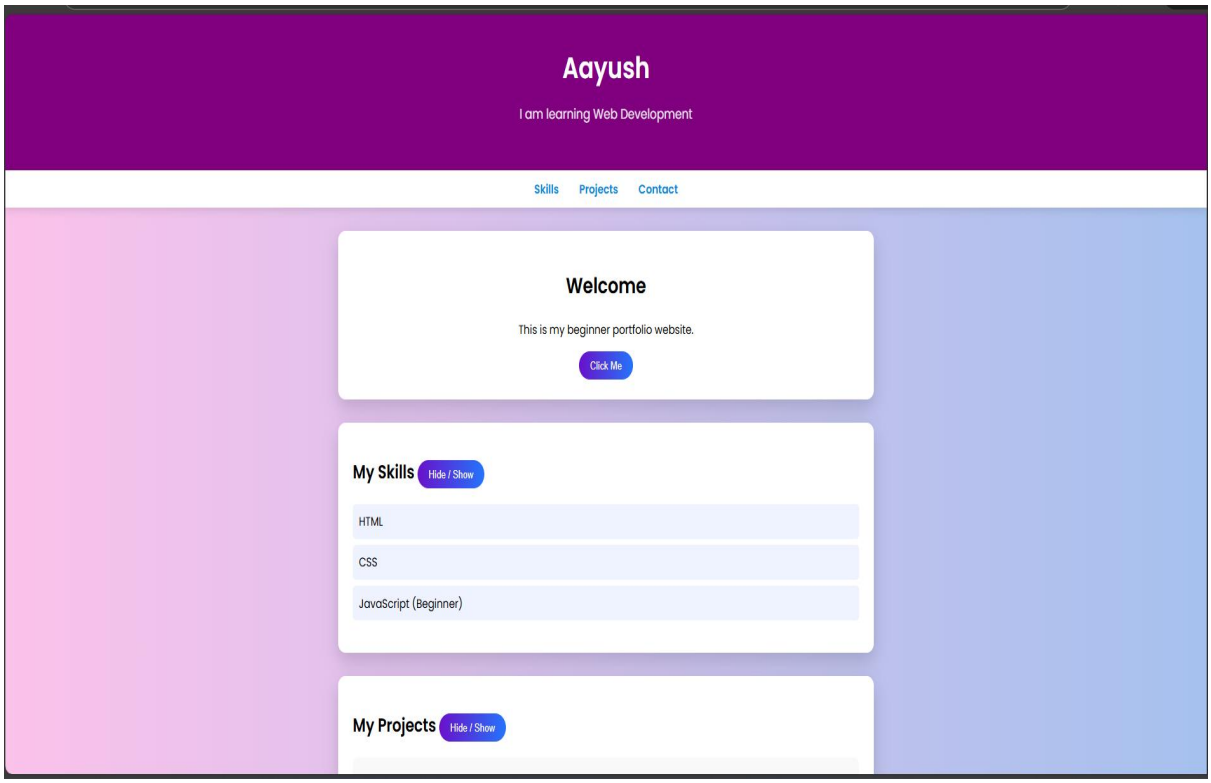
```
    border-color: #2575fc;
}
#contact a {
    display: inline-block;
    margin: 10px 10px 0 0;
    text-decoration: none;
    font-weight: 600;
    color: #2575fc;
}
#contact a:hover {
    color: #6a11cb;
}
footer {
    background-color: #111;
    color: white;
    text-align: center;
    padding: 15px;
    margin-top: 40px;
}
#scrollBtn {
    position: fixed;
    bottom: 25px;
    right: 25px;
    display: none;
    background: linear-gradient(to right, #6a11cb, #2575fc);
    color: white;
    border-radius: 50%;
    border: none;
    padding: 12px 16px;
    font-size: 18px;
    box-shadow: 0 8px 15px rgba(0,0,0,0.3);
```

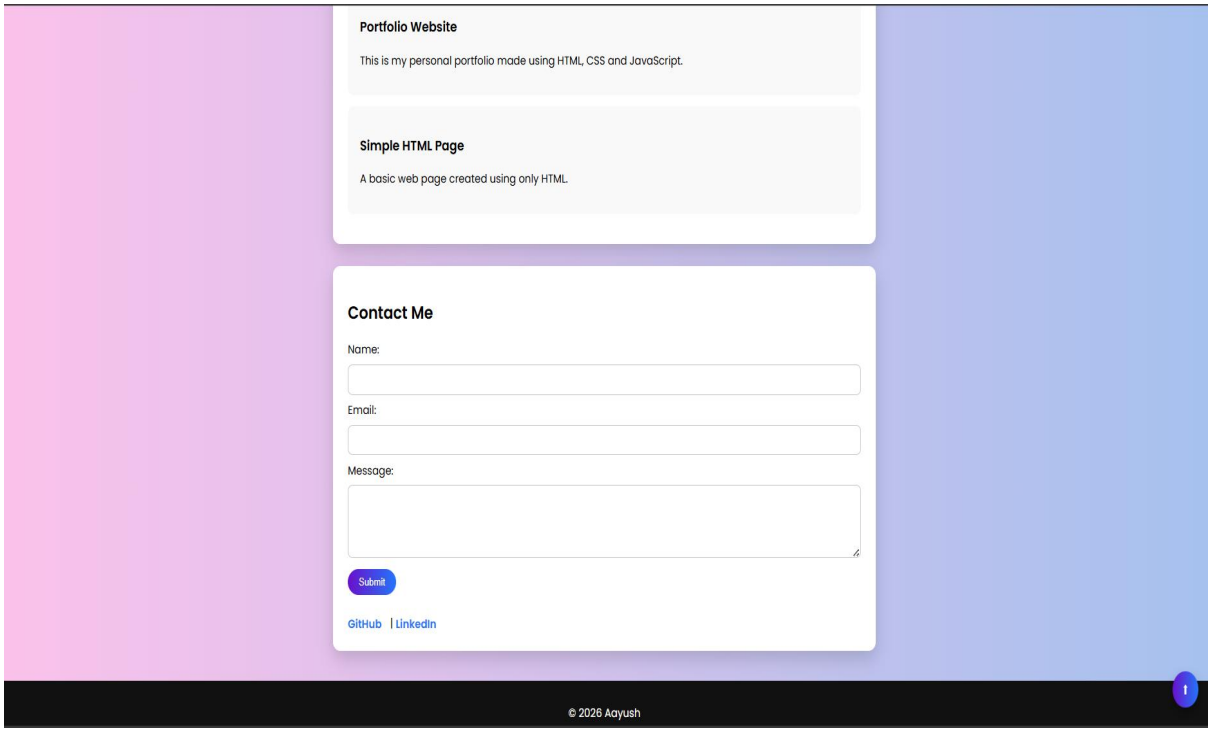
}

➤ **JS CODE :**

```
function showMessage() {
   alert("Thank you for visiting");
}
function showMessage() {
   alert("Welcome to my portfolio website!");
}
function submitForm() {
   document.getElementById("result").innerText =
      "Thank you! Your message has been submitted.";
}
function toggleSection(id) {
   const section = document.getElementById(id);
   if (section.style.display === "none") {
      section.style.display = "block";
   } else {
      section.style.display = "none";
   }
}
window.onscroll = function () {
   const btn = document.getElementById("scrollBtn");
   if (document.documentElement.scrollTop > 200) {
      btn.style.display = "block";
   } else {
      btn.style.display = "none";
   }
};
function scrollToTop() {
   window.scrollTo({ top: 0, behavior: "smooth" });
}
```

❖ OUTPUT:


Aayush
I am learning Web Development
Skills
Projects
Contact
Welcome
This is my beginner portfolio website.
Click Me
My Skills
Hide / Show
HTML
CSS
JavaScript (Beginner)
My Projects
Hide / Show


Portfolio Website
This is my personal portfolio made using HTML, CSS and JavaScript.
Simple HTML Page
A basic web page created using only HTML.
Contact Me
Name:
Email:
Message:
Submit
GitHub | LinkedIn
© 2026 Aayush

EXPERIMENT – II

❖ **Aim :**

Creating a **Company Landing Page** using **HTML and CSS** to display company information, services, testimonials, and contact details.

❖ **Tasks :**

- **Project Setup & Introduction** – Create project folder and basic HTML and CSS files.
- **HTML Structure** – Build landing page sections using semantic HTML tags.
- **Styling with CSS** – Apply layout, colors, fonts, spacing, and borders for a professional design.
- **Services Section Design** – Display services with service name and short description.
- **Testimonials Section** – Add customer feedback and style it neatly.
- **Contact Section (Bonus)** – Create contact information and a simple contact form.

❖ **Tech Stack Used :**

- **HTML , CSS and Javascript**

❖ **CODES :**

➤ **HTML CODE :**

```
<!--  -->

<!-- Name: Aayush

Project: TechNova Solutions -->

<!--  -->

<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <title>Company Landing Page</title>

   <link rel="stylesheet" href="style.css">

</head><wbr>


</wbr>
```

```
<body>
  <header>
    <h1>Aayush  Solutions</h1>
    <p>Innovating Technology for a Better Tomorrow</p>
  </header>
  <section>
    <h2>About Us</h2>
    <p>
      Aayush Solutions provides modern and reliable
      technology services for businesses and individuals.
    </p>
  </section>

  <section>
    <h2>Our Services</h2>

    <div class="service-card">
      <h3>Web Development</h3>
      <p>We create simple and responsive websites.</p>
    </div>
    <div class="service-card">
      <h3>App Development</h3>
      <p>We develop easy-to-use mobile applications.</p>
    </div>

    <div class="service-card">
      <h3>Technical Support</h3>
      <p>We provide 24/7 technical assistance.</p>
    </div>
  </section>
```

```
<section>
  <h2>Testimonials</h2>

  <div class="testimonial">
    <p>"Great service and friendly team!"</p>
    <h4>- Aayush</h4>
  </div>

  <div class="testimonial">
    <p>"Very professional and reliable company."</p>
    <h4>- Vansh</h4>
  </div>
</section>

<section>
  <h2>Contact Us</h2>

  <p>Email: Aayush@technova.com</p>
  <p>Phone: 9876543210</p>
  <p>Address: New Delhi, India</p>

  <form onsubmit="showMessage(); return false;">
    <input type="text" placeholder="Name" required><br><br>
    <input type="email" placeholder="Email" required><br><br>
    <textarea rows="4" placeholder="Message" required></textarea><br><br>
    <button type="submit">Send</button>
  </form>
</section>

<footer>
  <p>@ Aayush Solutions</p>
```

```
    </footer>

    <script src="script.js"></script>

</body>
</html>
```

➤ **CSS CODE :**

```
@import url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap');
* {
    box-sizing: border-box;
    font-family: 'Poppins', sans-serif;
}
body {
    margin: 0;
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: #333;
}
header {
    text-align: center;
    padding: 50px 20px;
    color: white;
}
header h1 {
    font-size: 42px;
    margin-bottom: 10px;
}
header p {
    font-size: 18px;
    opacity: 0.9;
}
```

```
section {
    background-color: white;
    max-width: 1000px;
    margin: 30px auto;
    padding: 30px;
    border-radius: 15px;
    box-shadow: 0 15px 30px rgba(0,0,0,0.2);
}
section h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #667eea;
}
.service-card {
    background: #f4f6ff;
    border-radius: 12px;
    padding: 20px;
    margin: 15px 0;
    transition: 0.3s ease;
}
.service-card h3 {
    margin-top: 0;
    color: #333;
}
.service-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 10px 20px rgba(0,0,0,0.2);
}
.testimonial {
    background: #f9f9f9;
    border-left: 5px solid #667eea;
```

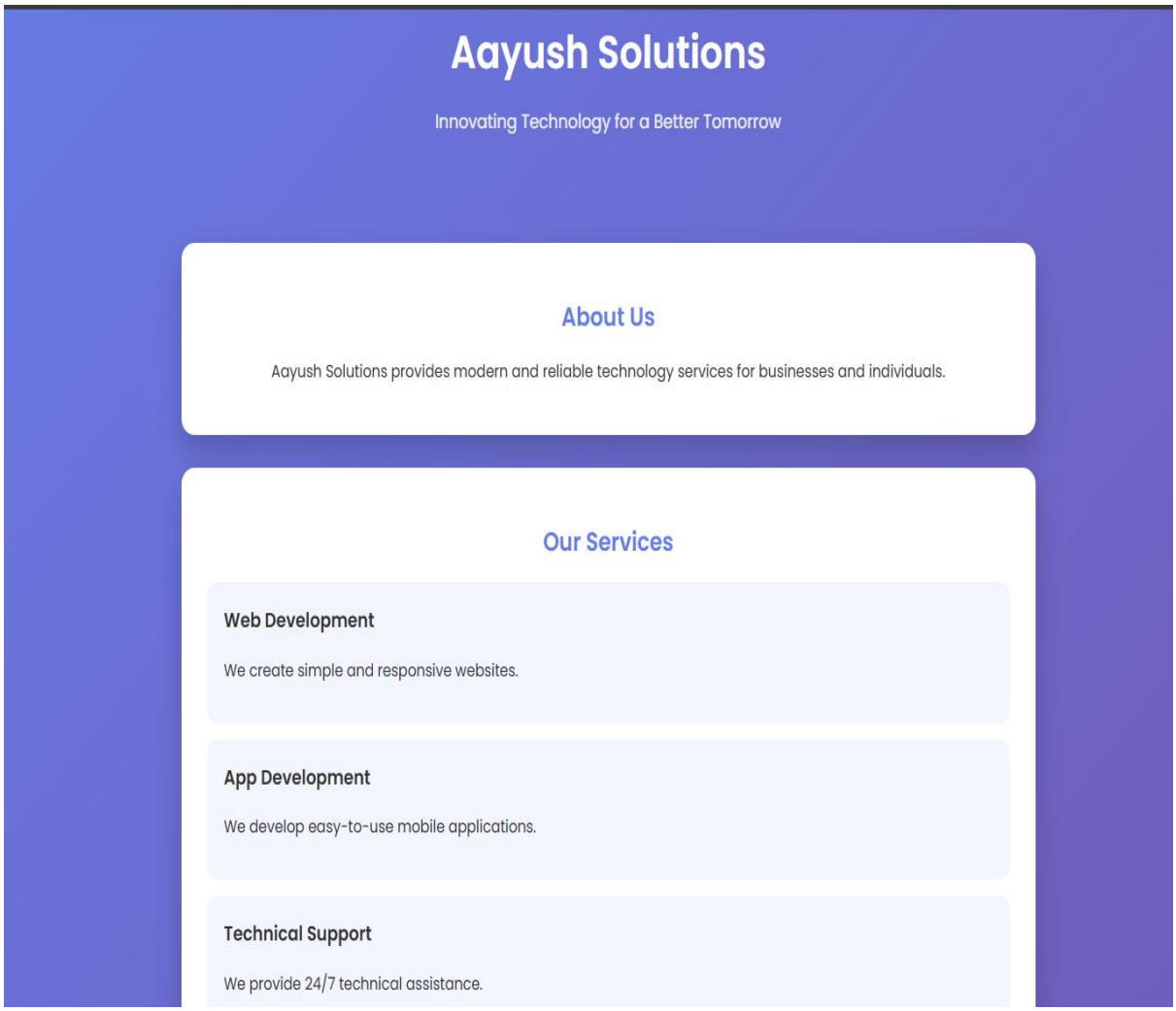
```
    padding: 15px 20px;
    margin: 15px 0;
    border-radius: 10px;
}
.testimonial p {
    font-style: italic;
}
.testimonial h4 {
    margin: 10px 0 0;
    text-align: right;
    color: #555;
}
section p {
    text-align: center;
}
form {
    max-width: 500px;
    margin: 20px auto 0;
    text-align: center;
}
input, textarea {
    width: 100%;
    padding: 12px;
    border-radius: 10px;
    border: 1px solid #ccc;
    font-size: 14px;
}
input:focus, textarea:focus {
    outline: none;
    border-color: #667eea;
}
```

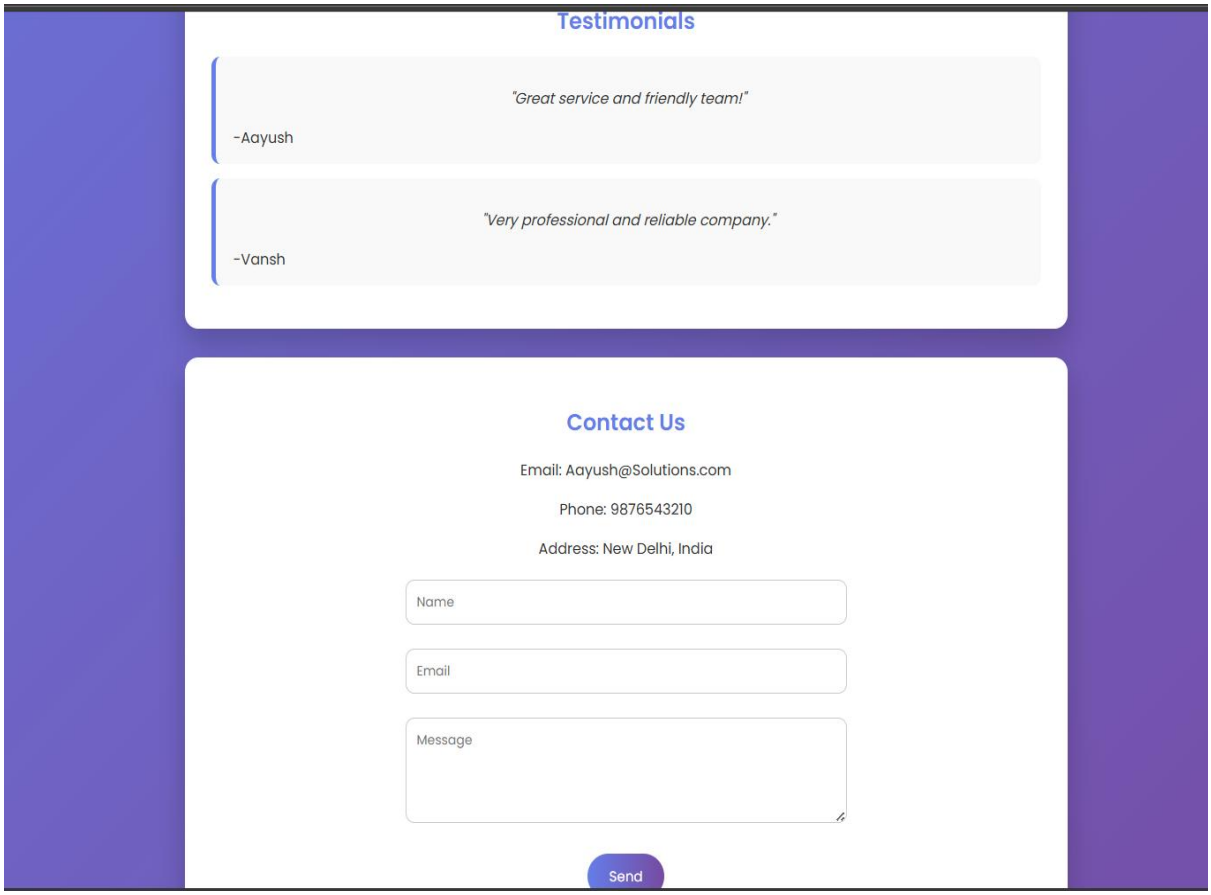
```
button {
    background: linear-gradient(to right, #667eea, #764ba2);
    color: white;
    border: none;
    padding: 12px 25px;
    border-radius: 25px;
    cursor: pointer;
    font-size: 15px;
    transition: 0.3s ease;
}
button:hover {
    transform: scale(1.05);
    box-shadow: 0 8px 20px rgba(0,0,0,0.3);
}
footer {
    text-align: center;
    padding: 15px;
    color: white;
    opacity: 0.9;
}
@media (max-width: 768px) {
    header h1 {
        font-size: 32px;
    }
    section {
        margin: 20px;
        padding: 25px;
    }
}
```

➤ **JS CODE :**

```
function showMessage() {
   alert("Thank you for contacting TechNova Solutions!");
}
```

❖ **OUTPUT :**




Testimonials
"Great service and friendly team!"
-Aayush
"Very professional and reliable company."
-Vansh
Contact Us
Email: Aayush@Solutions.com
Phone: 9876543210
Address: New Delhi, India
Name
Email
Message
Send

EXPERIMENT – 3

❖ **Aim :**
To design and develop an **Interactive Quiz Web Application** using **HTML, CSS, and JavaScript**.

❖ **Tasks :**
• **Project Setup & Introduction** – Create the project folder and link HTML, CSS, and JavaScript files properly.
• **Quiz Interface Design** – Design quiz layout with questions, options, and navigation buttons using semantic HTML.
• **Styling with CSS** – Apply colors, spacing, alignment, and layout to improve user experience.
• **Quiz Logic Implementation** – Store questions and answers in JavaScript and handle user input.
• **Score Tracking & Navigation** – Move between questions and calculate the score dynamically.

❖ **Tech Stack Used :**
• **HTML** •**CSS** • **JavaScript**

❖ Html Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Interactive Quiz</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
    <h1>Interactive Quiz</h1>
    <p>Test your Web Development knowledge</p>
</header>
```

```html
<main class="quiz-container">

    <section id="intro">
        <h2>Quiz Introduction</h2>
        <p>This quiz contains multiple-choice questions. Click Start to begin.</p>
        <button onclick="startQuiz()">Start Quiz</button>
    </section>

    <section id="quiz" class="hidden">
        <h2 id="question"></h2>
        <div id="options"></div>
        <button onclick="nextQuestion()">Next</button>
    </section>

    <section id="result" class="hidden">
        <h2>Your Result</h2>
        <p id="scoreText"></p>
        <p id="feedback"></p>
        <button onclick="restartQuiz()">Restart Quiz</button>
    </section>

</main>
<script src="script.js"></script>
</body>
</html>
```

➤ **CSS CODE :**

```css
* {
    box-sizing: border-box;
```

```
    font-family: Arial, sans-serif;
}

body {
    margin: 0;
    background: linear-gradient(to right, #74ebd5, #9face6);
}

header {
    background-color:blueviolet ;
    color: white;
    padding: 20px;
    text-align: center;
}

.quiz-container {
    background-color: white;
    max-width: 600px;
    margin: 30px auto;
    padding: 25px;
    border-radius: 10px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.2);
}

button {
    background-color: #3498db;
    color: white;
    border: none;
    padding: 10px 18px;
```

```
    margin-top: 15px;
    cursor: pointer;
    border-radius: 5px;
}

button:hover {
    background-color: #2980b9;
}

.option {
    background-color: #ecf0f1;
    padding: 10px;
    margin: 8px 0;
    border-radius: 5px;
    cursor: pointer;
}

.option:hover {
    background-color: #dcdde1;
}

.hidden {
    display: none;
}

footer {
    background-color: #2c3e50;
    color: white;
    text-align: center;
```

```
    padding: 10px;
}
```

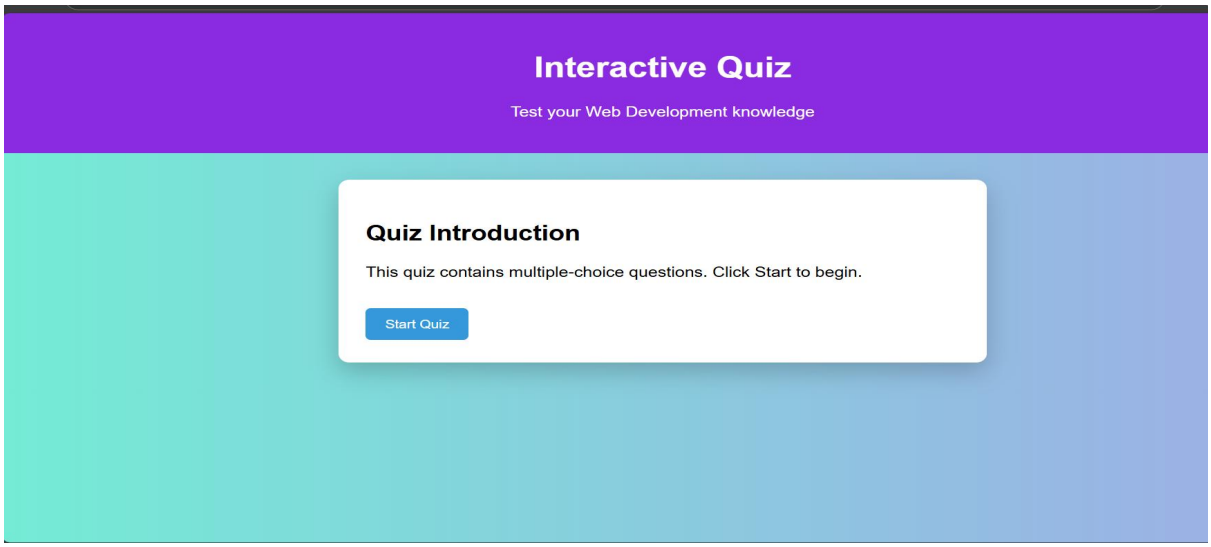
➤ **JS CODE :**

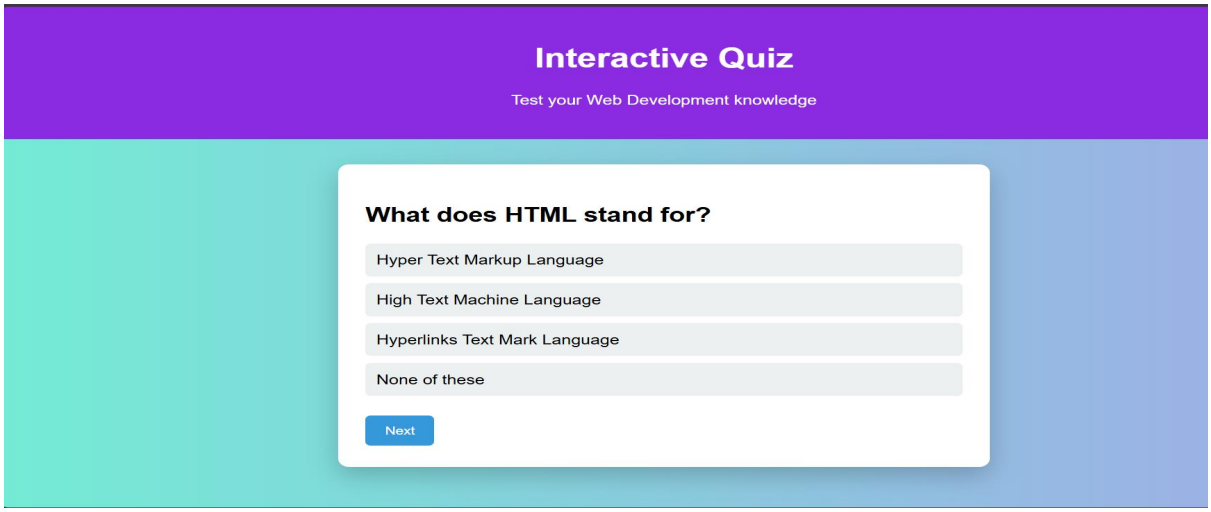
```
const quizData = [
    {
        question: "What does HTML stand for?",
        options: [
            "Hyper Text Markup Language",
            "High Text Machine Language",
            "Hyperlinks Text Mark Language",
            "None of these"
        ],
        correct: 0
    },
    {
        question: "Which language is used for styling web pages?",
        options: ["HTML", "CSS", "Java", "Python"],
        correct: 1
    },
    {
        question: "Which is used for web page interactivity?",
        options: ["HTML", "CSS", "JavaScript", "SQL"],
        correct: 2
    }
];

let currentQuestion = 0;
let score = 0;
```

```
function startQuiz() {
    document.getElementById("intro").classList.add("hidden");
    document.getElementById("quiz").classList.remove("hidden");
    loadQuestion();
}

function loadQuestion() {
    const q = quizData[currentQuestion];
    document.getElementById("question").innerText = q.question;

    const optionsDiv = document.getElementById("options");
    optionsDiv.innerHTML = "";

    q.options.forEach((option, index) => {
        const div = document.createElement("div");
        div.className = "option";
        div.innerText = option;
        div.onclick = () => selectAnswer(index);
        optionsDiv.appendChild(div);
    });
}

function selectAnswer(selected) {
    if (selected === quizData[currentQuestion].correct) {
        score++;
    }
    nextQuestion();
}
```

```
function nextQuestion() {
    currentQuestion++;
    if (currentQuestion < quizData.length) {
        loadQuestion();
    } else {
        showResult();
    }
}
function showResult() {
    document.getElementById("quiz").classList.add("hidden");
    document.getElementById("result").classList.remove("hidden");

    document.getElementById("scoreText").innerText =
        `Your Score: ${score} / ${quizData.length}`;
    let feedback = "Try Again";
    if (score === quizData.length) feedback = "Excellent";
    else if (score >= 2) feedback = "Good";
    document.getElementById("feedback").innerText = feedback;
}
function restartQuiz() {
    currentQuestion = 0;
    score = 0;
    document.getElementById("result").classList.add("hidden");
    document.getElementById("intro").classList.remove("hidden")
}
```

➤ Output :


Interactive Quiz
Test your Web Development knowledge
Quiz Introduction
This quiz contains multiple-choice questions. Click Start to begin.
Start Quiz


Interactive Quiz
Test your Web Development knowledge
What does HTML stand for?
Hyper Text Markup Language
High Text Machine Language
Hyperlinks Text Mark Language
None of these
Next

Interactive Quiz
Test your Web Development knowledge
Your Result
Your Score: 3 / 3
Excellent
Restart Quiz

Experiment : 4

❖ **Aim :**
To design and develop a **Dynamic To-Do List Web Application** using **HTML, CSS, and JavaScript** that allows users to add, edit, delete, and manage tasks interactively.

❖ **Tasks :**

• **Project Setup & Introduction** – Create a project folder named **dynamic_todo_list** and add linked HTML, CSS, and JavaScript files with a short description of the application.

• **To-Do List Interface Design** – Create an input field to enter tasks, an **Add Task** button, and display tasks in a list format using HTML.

• **Styling with CSS** – Apply CSS styling to input fields, buttons, and task list to ensure proper alignment, spacing, and visual clarity.

• **Add & Display Tasks Using JavaScript** – Capture user input, add tasks dynamically to the list, and prevent empty tasks from being added.

• **Edit & Delete Task Features** – Provide **Edit** and **Delete** buttons for each task to allow modification and removal of tasks dynamically.

• **Task Completion Status (Bonus)** – Add a checkbox or toggle to mark tasks as completed and visually indicate completed tasks using a strikethrough effect.

❖ **Tech Stack Used :**
• **HTML** • **CSS** • **JavaScript**

❖ Html Code :

```
<!--

Project Title: Dynamic To-Do List

Name: Aayush

Experiment: 4

Date: 2026

-->

<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">
```

```html
    <title>Dynamic To-Do List</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
    <h1>My To-Do List</h1>
    <p>A simple to-do list to add, edit, delete, and complete tasks</p>
</header>
<main class="container">
    <div class="input-area">
        <input type="text" id="taskInput" placeholder="Enter a new task">
        <button onclick="addTask()">Add Task</button>
    </div>

    <ul id="taskList">
    </ul>

</main>

<footer>
    <p>© 2026 Aayush</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

❖ **CSS Code :**

```
* {
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}
body {
    margin: 0;
    background-color: white;
}
header {
    background-color: blueviolet;
    color: white;
    padding: 20px;
    text-align: center;
}
.container {
    background-color: white;
    max-width: 500px;
    margin: 30px auto;
    padding: 20px;
    border-radius: 10px;
}

.input-area {
    display: flex;
    gap: 10px;
}
```

```
input {
    flex: 1;
    padding: 10px;
    font-size: 14px;
}

button {
    padding: 10px 15px;
    background-color: blue;
    color: white;
    border: none;
    cursor: pointer;
    border-radius: 5px;
}

button:hover {
    background-color: #3b55a0;
}

ul {
    list-style: none;
    padding: 0;
    margin-top: 20px;
}

li {
    background-color: whitesmoke;
    padding: 10px;
```

```
    margin-bottom: 10px;
    border-radius: 6px;
    display: flex;
    align-items: center;
    justify-content: space-between;
}

.completed {
    text-decoration: line-through;
    color: gray;
}

.task-buttons button {
    margin-left: 5px;
    font-size: 12px;
}

footer {
    text-align: center;
    padding: 10px;
    color: black;
}
```

❖**Javascript Code :**

```
function addTask() {
    const input = document.getElementById("taskInput");
    const taskText = input.value.trim();
```

```
if (taskText === "") {
    alert("Please enter a task");
    return;
}

const li = document.createElement("li");

const checkbox = document.createElement("input");
checkbox.type = "checkbox";
checkbox.onclick = () => toggleTask(li);

const span = document.createElement("span");
span.innerText = taskText;

const btnDiv = document.createElement("div");
btnDiv.className = "task-buttons";

const editBtn = document.createElement("button");
editBtn.innerText = "Edit";
editBtn.onclick = () => editTask(span);

const deleteBtn = document.createElement("button");
deleteBtn.innerText = "Delete";
deleteBtn.onclick = () => deleteTask(li);

btnDiv.appendChild(editBtn);
btnDiv.appendChild(deleteBtn);
```

```
    li.appendChild(checkbox);
    li.appendChild(span);
    li.appendChild(btnDiv);

    document.getElementById("taskList").appendChild(li);

    input.value = "";
}

function deleteTask(task) {
    task.remove();
}

function editTask(span) {
    const newText = prompt("Edit your task:", span.innerText);
    if (newText !== null && newText.trim() !== "") {
        span.innerText = newText;
    }
}

function toggleTask(task) {
    task.classList.toggle("completed");
}
```

❖**Server:**

```
require('dotenv').config();
const express = require('express');
```

```
const cors = require('cors');
const nodemailer = require('nodemailer');

const app = express();
app.use(cors());
app.use(express.json());

const SMTP_HOST = process.env.SMTP_HOST;
const SMTP_PORT = process.env.SMTP_PORT || 587;
const SMTP_USER = process.env.SMTP_USER;
const SMTP_PASS = process.env.SMTP_PASS;
const RECIPIENT_EMAIL = process.env.RECIPIENT_EMAIL;
const REMINDER_MINUTES = parseInt(process.env.REMINDER_MINUTES || '15', 10);

if (!SMTP_HOST || !SMTP_USER || !SMTP_PASS || !RECIPIENT_EMAIL) {
  console.warn('SMTP credentials or RECIPIENT_EMAIL not fully configured.
See .env.example');
}

let transporter;
async function createTransporter() {
  if (SMTP_HOST && SMTP_USER && SMTP_PASS) {
    return nodemailer.createTransport({
      host: SMTP_HOST,
      port: Number(SMTP_PORT),
      secure: Number(SMTP_PORT) === 465,
      auth: { user: SMTP_USER, pass: SMTP_PASS }
    });
```

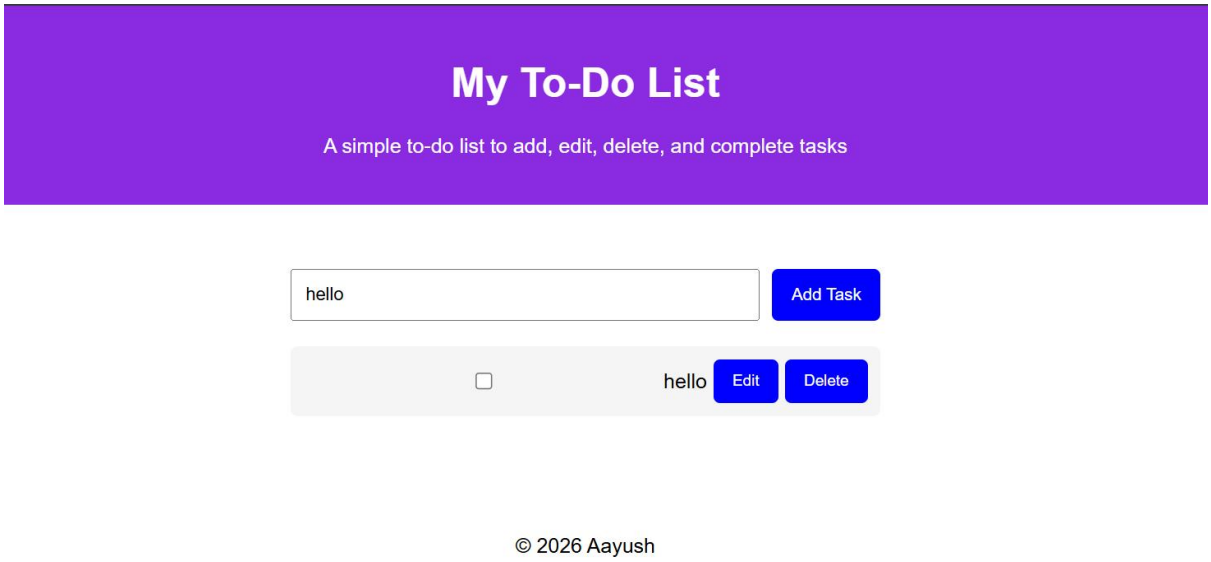
```
  }

  // Fallback: create a test account (Ethereal) for development/testing
  const testAccount = await nodemailer.createTestAccount();
  console.log('Using Nodemailer test account. Messages will be available at preview URLs.');
  return nodemailer.createTransport({
    host: 'smtp.ethereal.email',
    port: 587,
    auth: { user: testAccount.user, pass: testAccount.pass }
  });
}

(async () => { transporter = await createTransporter(); })();

// Simple in-memory store of scheduled timers (not persisted)
const scheduled = new Map();

app.get('/', (req, res) => {
  res.send('To-Do Reminder Server running');
});

// POST /api/tasks { text, due }
app.post('/api/tasks', (req, res) => {
  const { text, due } = req.body;

  if (!text) return res.status(400).json({ error: 'Task text required' });
```

```
if (!due) {
  // If no due time provided, send immediate notification
  sendReminderEmail(text, new Date())
    .then(() => res.json({ status: 'sent_immediate' }))
    .catch(err => res.status(500).json({ error: err.message }));
  return;
}

const dueDate = new Date(due);
if (isNaN(dueDate)) return res.status(400).json({ error: 'Invalid due date' });

const reminderTime = new Date(dueDate.getTime() - REMINDER_MINUTES * 60000);
const delay = reminderTime.getTime() - Date.now();

const id = Date.now() + Math.random().toString(36).slice(2, 8);

if (delay <= 0) {
  // reminder time passed — send immediately
  sendReminderEmail(text, dueDate)
    .then(() => res.json({ status: 'sent_immediate' }))
    .catch(err => res.status(500).json({ error: err.message }));
  return;
}

const timer = setTimeout(() => {
  sendReminderEmail(text, dueDate)
    .catch(err => console.error('Failed to send reminder:', err.message));
  scheduled.delete(id);
```

```
  }, delay);

  scheduled.set(id, { text, due: dueDate.toISOString(), timer });

  res.json({ status: 'scheduled', id, sendAt: reminderTime.toISOString() });
});

async function sendReminderEmail(taskText, dueDate) {
  if (!transporter || !RECIPIENT_EMAIL) return Promise.reject(new Error('Email not configured'));

  const formattedDue = dueDate ? new Date(dueDate).toLocaleString() : 'soon';
  const mailOptions = {
    from: SMTP_USER || 'no-reply@example.com',
    to: RECIPIENT_EMAIL,
    subject: `Reminder: upcoming task — ${taskText}`,
    text: `Reminder: Your task "${taskText}" is due at ${formattedDue}. This is an automated reminder sent ${REMINDER_MINUTES} minutes before.`
  };

  try {
    const info = await transporter.sendMail(mailOptions);
    // Log preview URL for test accounts (Ethereal)
    const preview = nodemailer.getTestMessageUrl(info);
    if (preview) console.log('Preview URL:', preview);
    return info;
  } catch (err) {
    return Promise.reject(err);
```

```
  }
}

// end of file

const PORT = process.env.PORT || 3000;
app.listen(PORT, () => {
  console.log(`Reminder server listening on port ${PORT}`);
});
```

❖**Output :**



Experiment : 5

❖ **Aim :**

To design and develop a Simple Blog Web Application using Flask that allows users to create, read, update, and delete blog posts.

❖ **Tasks :**

• **Project Setup & Introduction** – Create a project folder named simple_blog, install Flask, and create required files such as app.py, templates, and static folder.

• **Flask App Initialization** – Import Flask modules, initialize the application, and create a home route to display blog posts using a list or dictionary.

• **Create Blog Posts** – Create a form to add new blog posts with title and content, and store them in the application.

• **Read & Display Blog Posts** – Display all blog posts on the homepage with proper formatting and navigation links.

• **Update Blog Posts** – Add an edit option to update existing blog posts with pre-filled data.

• **Delete Blog Posts** – Provide a delete option to remove blog posts and update the list dynamically.

• **Basic Styling & Navigation (Bonus)** – Apply CSS for layout, fonts, and spacing, and add navigation links like Home and Create Post.

❖ **Tech Stack Used :**

• Python • Flask • HTML • CSS

❖ Html Code :

>Index.html :

```
{% extends "base.html" %}

{% block title %}New Post — The Blog{% endblock %}

{% block content %}

  <!-- Page Hero -->

  <div class="page-hero">
```

```
  <div class="eyebrow">Write</div>
  <h1>New Post</h1>
  <p>Share your thoughts with the world.</p>
</div>

<!-- Create Form -->
<div class="form-card">
  <form method="POST" action="{{ url_for('create') }}">

    <!-- Title Field -->
    <div class="form-group">
      <label for="title">Post Title</label>
      <input
        type="text"
        id="title"
        name="title"
        placeholder="Enter a compelling title..."
        required
        autofocus
      />
    </div>

    <!-- Content Field -->
    <div class="form-group">
      <label for="content">Content</label>
      <textarea
        id="content"
        name="content"
```

```
        placeholder="Write your post content here..."
        required
      ></textarea>
    </div>

    <!-- Actions -->
    <div class="form-actions">
      <button type="submit" class="btn btn-primary">Publish Post</button>
      <a href="{{ url_for('index') }}" class="btn btn-ghost">Cancel</a>
    </div>

  </form>
</div>

{% endblock %}
```

>Base.html :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>{% block title %}Simple Blog{% endblock %}</title>

  <!-- Google Fonts: Playfair Display + DM Sans -->
  <link rel="preconnect" href="https://fonts.googleapis.com" />
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
```

```
  <link
href="https://fonts.googleapis.com/css2?family=Playfair+Display:ital,wght@0,700;1,4
00&family=DM+Sans:wght@300;400;500&display=swap" rel="stylesheet" />


  <style>
    :root {
      --bg:         #0f0f0f;
      --surface:    #191919;
      --border:     #2a2a2a;
      --accent:     #c9a96e;       /* warm gold */
      --accent-dim: #8a6f44;
      --text:       #e8e0d5;
      --muted:      #7a7068;
      --danger:     #b85c5c;
      --radius:     10px;
      --max-w:      760px;
      --font-head:  'Playfair Display', Georgia, serif;
      --font-body:  'DM Sans', sans-serif;
    }
    *, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
    html { scroll-behavior: smooth; }


    body {
      background: var(--bg);
      color: var(--text);
      font-family: var(--font-body);
      font-weight: 300;
      min-height: 100vh;
```

```
  line-height: 1.7;
}
body::before {
  content: '';
  position: fixed;
  inset: 0;
  background-image: url("data:image/svg+xml,%3Csvg viewBox='0 0 200 200' xmlns='http://www.w3.org/2000/svg'%3E%3Cfilter id='noise'%3E%3CfeTurbulence type='fractalNoise' baseFrequency='0.9' numOctaves='4' stitchTiles='stitch'/%3E%3C/filter%3E%3Crect width='100%25' height='100%25' filter='url(%23noise)' opacity='0.03'/%3E%3C/svg%3E");
  pointer-events: none;
  z-index: 0;
}
header {
  position: sticky;
  top: 0;
  z-index: 100;
  background: rgba(15,15,15,0.92);
  backdrop-filter: blur(12px);
  border-bottom: 1px solid var(--border);
}

.nav-inner {
  max-width: var(--max-w);
  margin: 0 auto;
  padding: 0 24px;
  height: 64px;
  display: flex;
```

```
  align-items: center;
  justify-content: space-between;
}

.brand {
  font-family: var(--font-head);
  font-size: 1.35rem;
  color: var(--accent);
  text-decoration: none;
  letter-spacing: 0.02em;
}

.brand span { font-style: italic; color: var(--text); font-weight: 400; }

nav { display: flex; gap: 8px; align-items: center; }

nav a {
  color: var(--muted);
  text-decoration: none;
  font-size: 0.875rem;
  font-weight: 400;
  padding: 7px 14px;
  border-radius: 6px;
  transition: all 0.2s;
}

nav a:hover { color: var(--text); background: var(--surface); }
```

```
.btn-nav {
  background: var(--accent) !important;
  color: #0f0f0f !important;
  font-weight: 500 !important;
}

.btn-nav:hover { background: var(--accent-dim) !important; }
main {
  position: relative;
  z-index: 1;
  max-width: var(--max-w);
  margin: 0 auto;
  padding: 48px 24px 80px;
}
.page-hero {
  margin-bottom: 40px;
  padding-bottom: 32px;
  border-bottom: 1px solid var(--border);
}

.page-hero .eyebrow {
  font-size: 0.75rem;
  letter-spacing: 0.15em;
  text-transform: uppercase;
  color: var(--accent);
  margin-bottom: 10px;
}
```

```
.page-hero h1 {
  font-family: var(--font-head);
  font-size: clamp(2rem, 5vw, 3rem);
  line-height: 1.15;
  color: var(--text);
  font-weight: 700;
}

.page-hero p {
  margin-top: 12px;
  color: var(--muted);
  font-size: 0.95rem;
}

/* — Blog Post Cards ———————————————————————— */
.post-list { display: flex; flex-direction: column; gap: 20px; }

.post-card {
  background: var(--surface);
  border: 1px solid var(--border);
  border-radius: var(--radius);
  padding: 28px 32px;
  transition: border-color 0.25s, transform 0.2s;
  animation: fadeUp 0.4s ease both;
}

.post-card:hover { border-color: var(--accent-dim); transform: translateY(-2px); }
```

```
@keyframes fadeUp {
  from { opacity: 0; transform: translateY(16px); }
  to   { opacity: 1; transform: translateY(0); }
}

.post-meta {
  font-size: 0.75rem;
  letter-spacing: 0.08em;
  text-transform: uppercase;
  color: var(--accent);
  margin-bottom: 8px;
}

.post-card h2 {
  font-family: var(--font-head);
  font-size: 1.45rem;
  font-weight: 700;
  color: var(--text);
  margin-bottom: 10px;
  line-height: 1.3;
}

.post-card p {
  color: var(--muted);
  font-size: 0.92rem;
  display: -webkit-box;
  -webkit-line-clamp: 3;
  -webkit-box-orient: vertical;
```

```
  overflow: hidden;
}

.post-actions {
  display: flex;
  gap: 10px;
  margin-top: 22px;
  padding-top: 18px;
  border-top: 1px solid var(--border);
}

/* — Buttons ———————————————————————————————— */
.btn {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  padding: 8px 18px;
  border-radius: 6px;
  font-family: var(--font-body);
  font-size: 0.82rem;
  font-weight: 500;
  cursor: pointer;
  text-decoration: none;
  border: 1px solid transparent;
  transition: all 0.18s;
}

.btn-primary {
```

```
  background: var(--accent);
  color: #0f0f0f;
  border-color: var(--accent);
}

.btn-primary:hover { background: var(--accent-dim); border-color: var(--accent-dim); }

.btn-ghost {
  background: transparent;
  color: var(--muted);
  border-color: var(--border);
}

.btn-ghost:hover { color: var(--text); border-color: #444; background: var(--surface); }

.btn-danger {
  background: transparent;
  color: var(--danger);
  border-color: #3a2020;
}

.btn-danger:hover { background: #2a1515; border-color: var(--danger); }

/* — Empty State ———————————————————————————— */
.empty {
  text-align: center;
```

```
  padding: 64px 24px;
  color: var(--muted);
}

.empty .emoji { font-size: 3rem; margin-bottom: 16px; }
.empty h3 { font-family: var(--font-head); color: var(--text); margin-bottom: 8px; }

/* — Form Styles ———————————————————————————— */
.form-card {
  background: var(--surface);
  border: 1px solid var(--border);
  border-radius: var(--radius);
  padding: 36px 40px;
  animation: fadeUp 0.35s ease both;
}

.form-group { margin-bottom: 24px; }

label {
  display: block;
  font-size: 0.78rem;
  letter-spacing: 0.12em;
  text-transform: uppercase;
  color: var(--accent);
  margin-bottom: 8px;
  font-weight: 500;
}
```

```
input[type="text"],
textarea {
  width: 100%;
  background: var(--bg);
  border: 1px solid var(--border);
  border-radius: 7px;
  color: var(--text);
  font-family: var(--font-body);
  font-size: 0.95rem;
  padding: 12px 16px;
  outline: none;
  transition: border-color 0.2s;
  resize: vertical;
}

input[type="text"]:focus,
textarea:focus { border-color: var(--accent); }

textarea { min-height: 180px; line-height: 1.65; }

.form-actions { display: flex; gap: 12px; align-items: center; margin-top: 8px; }

footer {
  position: relative;
  z-index: 1;
  text-align: center;
  padding: 28px 24px;
  border-top: 1px solid var(--border);
```

```
      color: var(--muted);
      font-size: 0.8rem;
    }

    footer span { color: var(--accent); }

    /* — Responsive ———————————————————————————————— */
    @media (max-width: 520px) {
      .form-card { padding: 24px 20px; }
      .post-card  { padding: 22px 20px; }
    }
  </style>
</head>
<body>

  <!-- — Navigation — -->
  <header>
    <div class="nav-inner">
      <a href="{{ url_for('index') }}" class="brand">The<span>Blog</span></a>
      <nav>
        <a href="{{ url_for('index') }}">Home</a>
        <a href="{{ url_for('create') }}" class="btn-nav">+ New Post</a>
      </nav>
    </div>
  </header>

  <!-- — Page Content — -->
  <main>
```

```
  {% block content %}{% endblock %}
 </main>

 <!-- — Footer — -->
 <footer>
  Built with <span>Flask</span> &amp; Python  ·  Experiment-5
 </footer>

</body>
</html>
```

>Create.html :

```
 <!-- Page Hero -->
 <div class="page-hero">
  <div class="eyebrow">Write</div>
  <h1>New Post</h1>
  <p>Share your thoughts with the world.</p>
 </div>

 <!-- Create Form -->
 <div class="form-card">
  <form method="POST" action="{{ url_for('create') }}">

   <!-- Title Field -->
   <div class="form-group">
```

```
    <label for="title">Post Title</label>
    <input
      type="text"
      id="title"
      name="title"
      placeholder="Enter a compelling title..."
      required
      autofocus
    />
  </div>

  <!-- Content Field -->
  <div class="form-group">
    <label for="content">Content</label>
    <textarea
      id="content"
      name="content"
      placeholder="Write your post content here..."
      required
    ></textarea>
  </div>

  <!-- Actions -->
  <div class="form-actions">
    <button type="submit" class="btn btn-primary">Publish Post</button>
    <a href="{{ url_for('index') }}" class="btn btn-ghost">Cancel</a>
  </div>
</form>
```

```
  </div>

{% endblock %}
```

>edit.html :

```
  <!-- Page Hero -->
  <div class="page-hero">
    <div class="eyebrow">Edit Post #{{ post.id }}</div>
    <h1>Update Post</h1>
    <p>Make changes and save your updated post.</p>
  </div>

  <!-- Edit Form — pre-filled with existing post data -->
  <div class="form-card">
    <form method="POST" action="{{ url_for('edit', post_id=post.id) }}">

      <!-- Title Field (pre-filled) -->
      <div class="form-group">
        <label for="title">Post Title</label>
        <input
          type="text"
          id="title"
          name="title"
          value="{{ post.title }}"
          required
```

```
          autofocus
        />
      </div>

      <!-- Content Field (pre-filled) -->
      <div class="form-group">
        <label for="content">Content</label>
        <textarea
          id="content"
          name="content"
          required
        >{{ post.content }}</textarea>
      </div>

      <!-- Actions -->
      <div class="form-actions">
        <button type="submit" class="btn btn-primary">Save Changes</button>
        <a href="{{ url_for('index') }}" class="btn btn-ghost">Discard</a>
      </div>

    </form>
  </div>

{% endblock %}
```

❖ CSS Code :

```
nav a.active {
  color: var(--text);
  background: var(--surface);
}
```

❖ App.py Code :

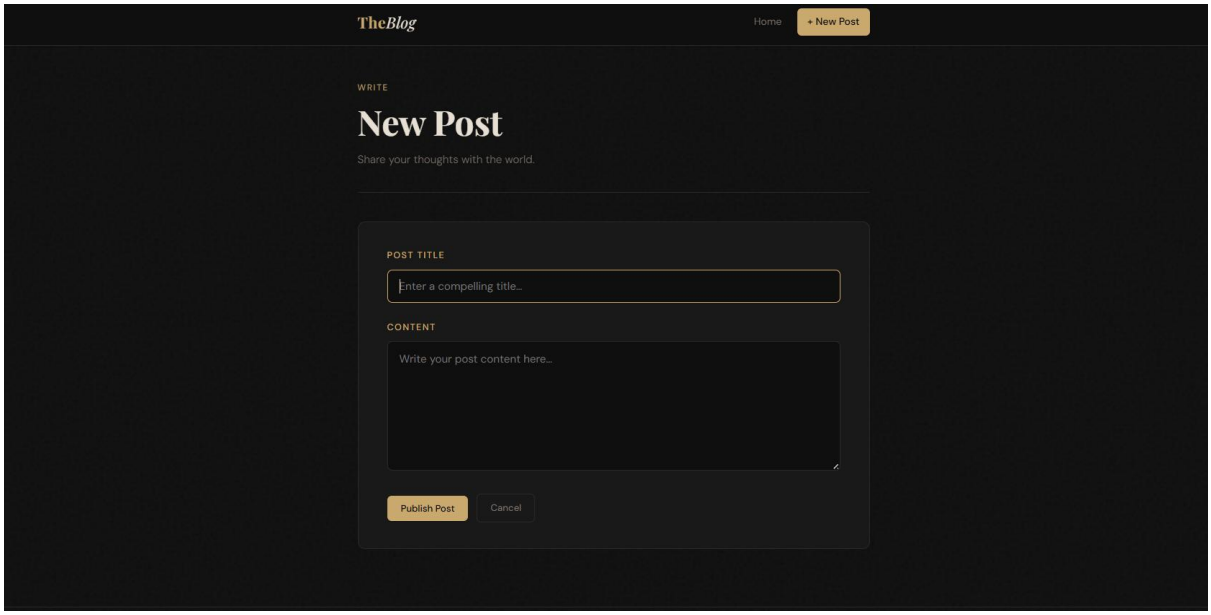
```
from flask import Flask, render_template, request, redirect, url_for
from datetime import datetime

# Initialize the Flask application
app = Flask(__name__)

# — In-memory storage for blog posts ————————————————
# Each post is a dictionary with: id, title, content, date
posts = [
    {
        "id": 1,
        "title": "Welcome to My Blog",
        "content": "This is the first post on this simple blog platform built with Flask. "
                   "You can create, edit, and delete posts easily!",
        "date": "April 01, 2026"
    },
    {
        "id": 2,
        "title": "Getting Started with Flask",
```

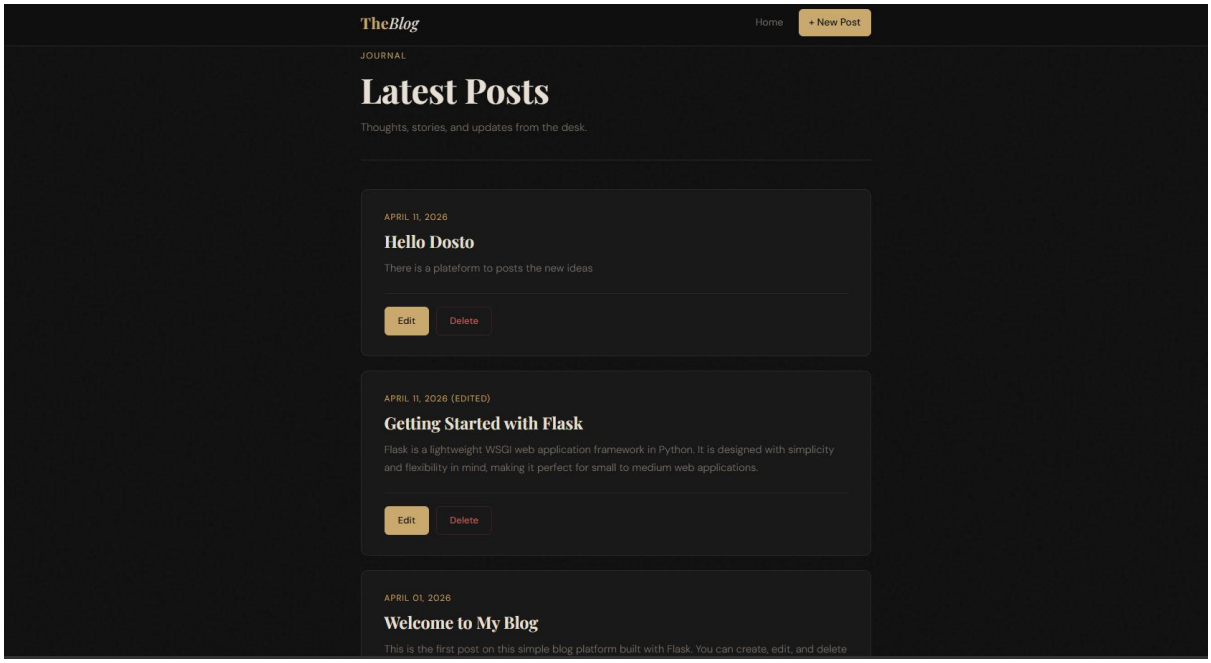
```
        "content": "Flask is a lightweight WSGI web application framework in Python. "
                   "It is designed with simplicity and flexibility in mind, making it "
                   "perfect for small to medium web applications.",
        "date": "April 02, 2026"
    }
]

# Counter to generate unique post IDs
next_id = 3

# — Helper: generate next unique ID ————————————————————
def get_next_id():
    global next_id
    current = next_id
    next_id += 1
    return current

@app.route("/")
def index():
    """Display all blog posts on the home page."""
    return render_template("index.html", posts=posts)

# — Route: Create Post (GET = show form, POST = save post) ——
@app.route("/create", methods=["GET", "POST"])
def create():
    """Handle creation of a new blog post."""
    if request.method == "POST":
        title   = request.form.get("title", "").strip()
```

```
        content = request.form.get("content", "").strip()

        # Basic validation
        if title and content:
            new_post = {
                "id":      get_next_id(),
                "title":   title,
                "content": content,
                "date":    datetime.now().strftime("%B %d, %Y")
            }
            posts.append(new_post)
            return redirect(url_for("index"))

    return render_template("create.html")

# — Route: Edit Post (GET = show pre-filled form, POST = update) —
@app.route("/edit/<int:post_id>", methods=["GET", "POST"])
def edit(post_id):
    """Handle editing an existing blog post."""
    # Find the post with matching ID
    post = next((p for p in posts if p["id"] == post_id), None)

    if post is None:
        return redirect(url_for("index"))   # Post not found → go home

    if request.method == "POST":
        title   = request.form.get("title", "").strip()
        content = request.form.get("content", "").strip()
```

```
        if title and content:
            post["title"]   = title
            post["content"] = content
            post["date"]    = datetime.now().strftime("%B %d, %Y") + " (edited)"

        return redirect(url_for("index"))

    return render_template("edit.html", post=post)

# — Route: Delete Post ————————————————————————————————
@app.route("/delete/<int:post_id>")
def delete(post_id):
    """Remove a blog post by its ID and redirect home."""
    global posts
    posts = [p for p in posts if p["id"] != post_id]
    return redirect(url_for("index"))

# — Entry Point ————————————————————————————————————
if __name__ == "__main__":
    app.run(debug=True)
```

❖ Output:


TheBlog
Home
+ New Post
WRITE
New Post
Share your thoughts with the world.
POST TITLE
Enter a compelling title...
CONTENT
Write your post content here...
Publish Post
Cancel


TheBlog
Home
+ New Post
JOURNAL
Latest Posts
Thoughts, stories, and updates from the desk.
APRIL 11, 2026
Hello Dosto
There is a plateform to posts the new ideas
Edit
Delete
APRIL 11, 2026 (EDITED)
Getting Started with Flask
Flask is a lightweight WSGI web application framework in Python. It is designed with simplicity and flexibility in mind, making it perfect for small to medium web applications.
Edit
Delete
APRIL 01, 2026
Welcome to My Blog
This is the first post on this simple blog platform built with Flask. You can create, edit, and delete

Experiment : 6

❖ **Aim :**

To design and develop a Contact Management System using Django that allows users to add, view, update, and delete contact details.

❖ **Tasks :**

• Project Setup & Introduction – Create a project folder named contact_management_system, set up a Django project and app, and create required folders such as templates and static.

• Django App Initialization – Configure the app in settings, create views, and set up URLs to display contacts on the homepage.

• Add New Contacts – Create a form to add new contacts with Name, Phone Number, and Email, and store data in the database.

• View & Display Contacts – Display all contacts in a table or list using Django templates with proper formatting and navigation.

• Update Contact Details – Provide an edit option to update existing contact details with pre-filled form data.

• Delete Contacts – Add a delete option to remove contacts from the database and update the list dynamically.

• Search & Styling (Bonus) – Add search functionality by name or phone number and apply basic CSS for better UI design.

❖ **Tech Stack Used :**

• Python • Django • HTML • CSS • SQLite

❖ Html Code :

>Index.html :

```
<div class="page-header d-flex flex-wrap align-items-center justify-content-between gap-3 mb-4">
  <div>
    <h1 class="page-title mb-1">
      <i class="bi bi-people-fill me-2"></i>My Contacts
```

```
    </h1>
    <p class="text-muted mb-0">
      {% if query %}
        Showing <strong>{{ result_count }}</strong> result(s) for "<em>{{ query }}</em>"
      {% else %}
        You have <strong>{{ total_contacts }}</strong> contact{{ total_contacts|pluralize }}
saved.
      {% endif %}
    </p>
  </div>
  <a href="{% url 'add_contact' %}" class="btn btn-primary btn-lg shadow">
    <i class="bi bi-person-plus-fill me-2"></i>Add New Contact
  </a>
</div>

<!-- ——— SEARCH BAR ——— -->
<!-- Sends a GET request to the same page with ?q=search_term -->
<div class="card search-card mb-4 shadow-sm">
  <div class="card-body py-3">
    <form method="GET" action="{% url 'index' %}" class="d-flex gap-2">
      <div class="input-group">
        <span class="input-group-text bg-white border-end-0">
          <i class="bi bi-search text-muted"></i>
        </span>
        <input
          type="text"
          name="q"
          value="{{ query }}"
          class="form-control border-start-0 ps-0"
          placeholder="Search by name, phone, or email..."
```

```
            autocomplete="off"
          >
        </div>
        <button type="submit" class="btn btn-primary px-4">
          Search
        </button>
        {% if query %}
        <a href="{% url 'index' %}" class="btn btn-outline-secondary px-3" title="Clear search">
          <i class="bi bi-x-lg"></i>
        </a>
        {% endif %}
      </form>
    </div>
</div>

<!-- ——— CONTACTS TABLE ——— -->
{% if contacts %}
<div class="card shadow-sm">
  <div class="card-body p-0">
    <div class="table-responsive">
      <table class="table table-hover contacts-table mb-0">
        <thead>
          <tr>
            <th>#</th>
            <th><i class="bi bi-person me-1"></i>Name</th>
            <th><i class="bi bi-telephone me-1"></i>Phone</th>
            <th><i class="bi bi-envelope me-1"></i>Email</th>
            <th><i class="bi bi-geo-alt me-1"></i>Address</th>
            <th class="text-center">Actions</th>
```

```
    </tr>
  </thead>
  <tbody>
    <!-- Loop through each contact — Django template for loop -->
    {% for contact in contacts %}
    <tr>
      <!-- forloop.counter gives 1, 2, 3, ... automatically -->
      <td class="text-muted">{{ forloop.counter }}</td>

      <td>
        <!-- Avatar circle with first letter of name -->
        <div class="d-flex align-items-center gap-2">
          <div class="avatar-circle">
            {{ contact.name|first|upper }}
          </div>
          <a href="{% url 'contact_detail' contact.pk %}" class="contact-name-link">
            {{ contact.name }}
          </a>
        </div>
      </td>

      <td>
        <a href="tel:{{ contact.phone }}" class="text-decoration-none text-dark">
          <i class="bi bi-telephone-fill text-success me-1"></i>
          {{ contact.phone }}
        </a>
      </td>

      <td>
```

```
    <a href="mailto:{{ contact.email }}" class="text-decoration-none text-dark">
      {{ contact.email }}
    </a>
  </td>

  <td class="text-muted">
    {% if contact.address %}
      {{ contact.address|truncatechars:30 }}
    {% else %}
      <span class="text-muted fst-italic">—</span>
    {% endif %}
  </td>

  <td class="text-center">
    <div class="action-btns d-flex gap-1 justify-content-center">
      <!-- View button -->
      <a href="{% url 'contact_detail' contact.pk %}"
        class="btn btn-sm btn-outline-info" title="View Details">
         <i class="bi bi-eye"></i>
      </a>
      <!-- Edit button -->
      <a href="{% url 'edit_contact' contact.pk %}"
        class="btn btn-sm btn-outline-warning" title="Edit">
         <i class="bi bi-pencil-square"></i>
      </a>
      <!-- Delete button — links to delete confirmation page -->
      <a href="{% url 'delete_contact' contact.pk %}"
        class="btn btn-sm btn-outline-danger" title="Delete">
         <i class="bi bi-trash3"></i>
```

```
                    </a>
                  </div>
                </td>
              </tr>
              {% endfor %}
            </tbody>
          </table>
        </div>
      </div>
    </div>

<!-- ——— EMPTY STATE — No search results ——— -->
{% else %}
<div class="empty-state text-center py-5">
  {% if query %}
    <div class="empty-icon mb-3">🔍</div>
    <h4>No contacts found for "<em>{{ query }}</em>"</h4>
    <p class="text-muted">Try searching with a different name or phone number.</p>
    <a href="{% url 'index' %}" class="btn btn-outline-secondary mt-2">
      <i class="bi bi-arrow-left me-1"></i> Back to All Contacts
    </a>
  {% else %}
    <div class="empty-icon mb-3">📭</div>
    <h4>No contacts yet!</h4>
    <p class="text-muted">Start building your contact list by adding your first contact.</p>
    <a href="{% url 'add_contact' %}" class="btn btn-primary mt-2">
      <i class="bi bi-person-plus-fill me-2"></i>Add Your First Contact
    </a>
  {% endif %}
```

```
</div>
{% endif %}
{% endblock %}
```

>edit_contact.html :

```
<nav aria-label="breadcrumb" class="mb-3">
    <ol class="breadcrumb">
        <li class="breadcrumb-item">
            <a href="{% url 'index' %}"><i class="bi bi-house-door me-
1"></i>Home</a>
        </li>
        <li class="breadcrumb-item active">Edit — {{ contact.name }}</li>
    </ol>
</nav>

<div class="row justify-content-center">
    <div class="col-lg-7 col-md-9">
        <div class="card shadow form-card">
            <div class="card-header form-card-header edit-header">
                <h4 class="mb-0">
                    <i class="bi bi-pencil-square me-2"></i>Edit Contact
                </h4>
                <p class="mb-0 mt-1 opacity-75 small">
                    Editing: <strong>{{ contact.name }}</strong>
                </p>
            </div>
```

```
<div class="card-body p-4">
    <form method="post" novalidate>
        {% csrf_token %}

        {% for field in form %}
        <div class="mb-3">
            <label for="{{ field.id_for_label }}" class="form-label fw-semibold">
                {{ field.label }}
                {% if field.field.required %}
                <span class="text-danger">*</span>
                {% endif %}
            </label>

            <!-- Fields are pre-filled because we passed instance=contact in
views.py -->
            {{ field }}

            {% if field.errors %}
                {% for error in field.errors %}
                <div class="invalid-feedback d-block">
                    <i class="bi bi-exclamation-circle me-1"></i>{{ error }}
                </div>
                {% endfor %}
                <script>
                    document.getElementById('{{ field.id_for_label }}').classList.add('is
-invalid');
                </script>
            {% endif %}
```

```
                </div>
                {% endfor %}

                <div class="d-flex gap-3 mt-4">
                    <button type="submit" class="btn btn-warning btn-lg flex-fill text-
white">
                        <i class="bi bi-check2-circle me-2"></i>Update Contact
                    </button>
                    <a href="{% url 'index' %}" class="btn btn-outline-secondary btn-lg
flex-fill">
                        <i class="bi bi-x-circle me-2"></i>Cancel
                    </a>
                </div>

            </form>
        </div>
    </div>

    <!-- Danger zone — quick delete link -->
    <div class="text-center mt-3">
        <a href="{% url 'delete_contact' contact.pk %}" class="text-danger small">
            <i class="bi bi-trash3 me-1"></i>Delete this contact instead
        </a>
    </div>

  </div>
</div>
```

```
{% endblock %}
```

>delete_confirm.html :

```
<div class="row justify-content-center">
    <div class="col-md-6">
        <div class="card shadow border-danger">
            <div class="card-header bg-danger text-white text-center py-4">
                <div style="font-size: 3rem;">⚠</div>
                <h4 class="mb-0 mt-2">Confirm Deletion</h4>
            </div>
            <div class="card-body text-center p-4">
                <p class="lead mb-1">Are you sure you want to delete</p>
                <h5 class="text-danger fw-bold mb-3">"{{ contact.name }}"?</h5>
                <p class="text-muted mb-4">
                    <i class="bi bi-exclamation-triangle me-1"></i>
                    This action <strong>cannot be undone</strong>.
                    The contact and all their information will be permanently removed.
                </p>

                <!-- Contact preview -->
                <div class="contact-preview-box p-3 mb-4 rounded text-start">
                    <p class="mb-1"><i class="bi bi-person me-2 text-
muted"></i><strong>{{ contact.name }}</strong></p>
                    <p class="mb-1"><i class="bi bi-telephone me-2 text-
muted"></i>{{ contact.phone }}</p>
                    <p class="mb-0"><i class="bi bi-envelope me-2 text-
muted"></i>{{ contact.email }}</p>
                </div>
```

```
            <div class="d-flex gap-3">
                <!-- POST form — actual delete happens here -->
                <!--
                    We use POST (not a simple link) for delete because:
                    1. GET requests should not change data (REST principle)
                    2. Prevents accidental deletion by visiting the URL
                -->
                <form method="post" class="flex-fill">
                    {% csrf_token %}
                    <button type="submit" class="btn btn-danger btn-lg w-100">
                        <i class="bi bi-trash3-fill me-2"></i>Yes, Delete
                    </button>
                </form>

                <!-- Cancel — go back home -->
                <a href="{% url 'index' %}" class="btn btn-outline-secondary btn-lg flex-
fill">
                    <i class="bi bi-arrow-left me-2"></i>Cancel
                </a>
            </div>

        </div>
    </div>
  </div>
</div>

{% endblock %}
```

>contact_detail.html :

```
<nav aria-label="breadcrumb" class="mb-3">
    <ol class="breadcrumb">
        <li class="breadcrumb-item">
            <a href="{% url 'index' %}"><i class="bi bi-house-door me-
1"></i>Home</a>
        </li>
        <li class="breadcrumb-item active">{{ contact.name }}</li>
    </ol>
</nav>

<div class="row justify-content-center">
    <div class="col-lg-6 col-md-8">
        <div class="card shadow detail-card">
            <!-- Large avatar at top -->
            <div class="card-header detail-header text-center py-4">
                <div class="detail-avatar mx-auto mb-3">
                    {{ contact.name|first|upper }}
                </div>
                <h3 class="mb-1 fw-bold">{{ contact.name }}</h3>
                <small class="opacity-75">
                    Added on {{ contact.created_at|date:"d M Y" }}
                </small>
```

```
        </div>


        <div class="card-body p-4">
            <!-- Contact details list -->
            <ul class="list-unstyled detail-list">
                <li class="detail-item">
                    <span class="detail-icon"><i class="bi bi-telephone-fill
text-success"></i></span>
                    <div>
                        <div class="detail-label">Phone Number</div>
                        <a href="tel:{{ contact.phone }}" class="detail-value
text-decoration-none">
                            {{ contact.phone }}
                        </a>
                    </div>
                </li>
                <li class="detail-item">
                    <span class="detail-icon"><i class="bi bi-envelope-fill
text-primary"></i></span>
                    <div>
                        <div class="detail-label">Email Address</div>
                        <a href="mailto:{{ contact.email }}" class="detail-value
text-decoration-none">
                            {{ contact.email }}
                        </a>
                    </div>
```

```
                </li>
                {% if contact.address %}
                <li class="detail-item">
                    <span class="detail-icon"><i class="bi bi-geo-alt-fill text-
danger"></i></span>
                    <div>
                        <div class="detail-label">Address</div>
                        <div class="detail-value">{{ contact.address }}</div>
                    </div>
                </li>
                {% endif %}
                <li class="detail-item">
                    <span class="detail-icon"><i class="bi bi-clock-fill text-
warning"></i></span>
                    <div>
                        <div class="detail-label">Last Updated</div>
                        <div class="detail-value">{{ contact.updated_at|date:"d
M Y, h:i A" }}</div>
                    </div>
                </li>
            </ul>

            <!-- Action buttons -->
            <div class="d-flex gap-3 mt-4">
                <a href="{% url 'edit_contact' contact.pk %}" class="btn btn-
warning btn-lg flex-fill text-white">
```

```
                <i class="bi bi-pencil-square me-2"></i>Edit
            </a>
            <a href="{% url 'delete_contact' contact.pk %}" class="btn
btn-outline-danger btn-lg flex-fill">
                <i class="bi bi-trash3 me-2"></i>Delete
            </a>
        </div>

        <div class="text-center mt-3">
            <a href="{% url 'index' %}" class="text-muted small">
                <i class="bi bi-arrow-left me-1"></i>Back to All Contacts
            </a>
        </div>
      </div>
    </div>
  </div>
</div>

{% endblock %}
```

>base.html :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{% block title %}Contact Management System{% endblock %}</title>

    <!-- Google Fonts — Poppins for headings, Nunito for body text -->
    <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600;700&family=Nunito:wght@400;500;600&display=swap" rel="stylesheet">

    <!-- Bootstrap 5 CSS — for layout and components -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">

    <!-- Bootstrap Icons -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.0/font/bootstrap-icons.css" rel="stylesheet">

    <!-- Our custom CSS -->
    {% load static %}
    <link rel="stylesheet" href="{% static 'contacts/css/style.css' %}">
</head>
<body>

    <!-- ——— NAVBAR ——— -->
```

```
<nav class="navbar navbar-expand-lg navbar-dark sticky-top">
    <div class="container">
        <!-- Brand / Logo -->
        <a class="navbar-brand d-flex align-items-center gap-2"
href="{% url 'index' %}">
            <span class="brand-icon">[redacted]</span>
            <span class="brand-text">ContactBook</span>
        </a>

        <!-- Mobile toggle button -->
        <button class="navbar-toggler" type="button" data-bs-
toggle="collapse" data-bs-target="#navbarNav">
            <span class="navbar-toggler-icon"></span>
        </button>

        <!-- Nav links -->
        <div class="collapse navbar-collapse" id="navbarNav">
            <ul class="navbar-nav ms-auto align-items-center gap-2">
                <li class="nav-item">
                    <a class="nav-link {% if request.resolver_match.url_name
== 'index' %}active{% endif %}"
                       href="{% url 'index' %}">
                        <i class="bi bi-house-door me-1"></i> Home
                    </a>
                </li>
                <li class="nav-item">
```

```
                    <a class="btn btn-add nav-link" href="{% url
'add_contact' %}">
                        <i class="bi bi-person-plus-fill me-1"></i> Add
Contact
                    </a>
                </li>
            </ul>
        </div>
    </div>
  </nav>
  <!-- Django messages are like notifications — shown once, then gone
-->
  {% if messages %}
  <div class="container mt-3">
    {% for message in messages %}
    <div class="alert alert-{{ message.tags }} alert-dismissible fade show
shadow-sm" role="alert">
        {{ message }}
        <button type="button" class="btn-close" data-bs-
dismiss="alert"></button>
    </div>
    {% endfor %}
  </div>
  {% endif %}


  <!-- ——— MAIN CONTENT ——— -->
```

```
<main class="container py-4">
    {% block content %}
    <!-- Each page fills this block -->
    {% endblock %}
</main>

<!-- ——— FOOTER ——— -->
<footer class="footer mt-auto py-3">
    <div class="container text-center">
        <p class="mb-0">
            <span class="footer-icon">📇</span>
            <strong>ContactBook</strong> &mdash; Contact Management System &mdash;
            Made with <span style="color:#e74c3c;">❤</span> using Django
        </p>
        <small class="text-muted">Experiment 6 &bull; Python Web Development</small>
    </div>
</footer>

<!-- Bootstrap 5 JS -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
```

```
    <!-- Auto-dismiss flash messages after 4 seconds -->
    <script>
        setTimeout(function () {
            let alerts = document.querySelectorAll('.alert');
            alerts.forEach(function (alert) {
                let bsAlert = new bootstrap.Alert(alert);
                bsAlert.close();
            });
        }, 4000);
    </script>

    {% block extra_js %}{% endblock %}
</body>
</html>
```

>add_contact.html :

```
<nav aria-label="breadcrumb" class="mb-3">
    <ol class="breadcrumb">
        <li class="breadcrumb-item">
            <a href="{% url 'index' %}"><i class="bi bi-house-door me-
1"></i>Home</a>
        </li>
        <li class="breadcrumb-item active">Add New Contact</li>
    </ol>
</nav>
```

```
<!-- Form card -->
<div class="row justify-content-center">
    <div class="col-lg-7 col-md-9">
        <div class="card shadow form-card">
            <div class="card-header form-card-header">
                <h4 class="mb-0">
                    <i class="bi bi-person-plus-fill me-2"></i>Add New
Contact
                </h4>
                <p class="mb-0 mt-1 opacity-75 small">Fill in the details
below to save a new contact.</p>
            </div>

            <div class="card-body p-4">
                <!--
                    method="post" — form data is sent as POST request
                    {% csrf_token %} — REQUIRED by Django for security
(Cross-Site Request Forgery protection)
                    Without csrf_token, Django will reject the form submission!
                -->
                <form method="post" novalidate>
                    {% csrf_token %}

                    <!-- Loop through each field in the form -->
                    {% for field in form %}
```

```
<div class="mb-3">
    <!-- Field label -->
    <label for="{{ field.id_for_label }}" class="form-label fw-semibold">
        {{ field.label }}
        <!-- Show a red * for required fields -->
        {% if field.field.required %}
        <span class="text-danger">*</span>
        {% endif %}
    </label>

    <!-- Render the input widget — Django generates the HTML -->
    {{ field }}

    <!-- Show validation errors in red -->
    {% if field.errors %}
        {% for error in field.errors %}
        <div class="invalid-feedback d-block">
            <i class="bi bi-exclamation-circle me-1"></i>{{ error }}
        </div>
        {% endfor %}
        <!-- Add Bootstrap's is-invalid class when there's an error -->
        <script>
```

```
                    document.getElementById('{{ field.id_for_label }}').classList.add('is-invalid');
                </script>
            {% endif %}

            <!-- Help text (if any) -->
            {% if field.help_text %}
            <div class="form-text text-muted">{{ field.help_text }}</div>
            {% endif %}
        </div>
        {% endfor %}

        <!-- Form buttons -->
        <div class="d-flex gap-3 mt-4">
            <button type="submit" class="btn btn-primary btn-lg flex-fill">
                <i class="bi bi-save2 me-2"></i>Save Contact
            </button>
            <a href="{% url 'index' %}" class="btn btn-outline-secondary btn-lg flex-fill">
                <i class="bi bi-x-circle me-2"></i>Cancel
            </a>
        </div>

    </form>
```

```
            </div><!-- /card-body -->
        </div><!-- /card -->

        <!-- Small hint -->
        <p class="text-muted text-center mt-3 small">
            <i class="bi bi-info-circle me-1"></i>
            Fields marked with <span class="text-danger">*</span> are required.
        </p>

    </div><!-- /col -->
</div><!-- /row -->

{% endblock %}
```

❖ **CSS Code :**

```
:root {
    --primary: #2563eb;
    --primary-dark: #1d4ed8;
    --primary-light: #dbeafe;
    --warning: #f59e0b;
    --danger: #ef4444;
    --success: #10b981;
    --bg: #f1f5f9;
    --card-bg: #ffffff;
```

```
    --text: #1e293b;
    --text-muted: #64748b;
    --border: #e2e8f0;
    --navbar-bg: #1e293b;
    --avatar-bg: #2563eb;
    --avatar-size: 38px;
    --radius: 12px;
}
body {
    font-family: 'Nunito', sans-serif;
    background-color: var(--bg);
    color: var(--text);
    min-height: 100vh;
    display: flex;
    flex-direction: column;
}

h1, h2, h3, h4, h5 {
    font-family: 'Poppins', sans-serif;
    font-weight: 600;
}
.navbar {
    background: var(--navbar-bg) !important;
    box-shadow: 0 2px 12px rgba(0,0,0,0.15);
    padding: 0.75rem 0;
```

```
}

.brand-icon {
    font-size: 1.4rem;
}

.brand-text {
    font-family: 'Poppins', sans-serif;
    font-size: 1.25rem;
    font-weight: 700;
    letter-spacing: 0.5px;
}

.nav-link {
    font-weight: 600;
    font-size: 0.95rem;
    transition: color 0.2s;
}

.navbar .nav-link.active {
    color: #93c5fd !important;
}

/* Add contact button in navbar */
.btn-add {
```

```
    background: var(--primary) !important;
    border-color: var(--primary) !important;
    color: white !important;
    border-radius: 8px !important;
    padding: 0.4rem 1rem !important;
    transition: background 0.2s, transform 0.1s;
}

.btn-add:hover {
    background: var(--primary-dark) !important;
    transform: translateY(-1px);
}
.page-title {
    font-size: 1.75rem;
    color: var(--text);
}
.search-card {
    border: 1px solid var(--border);
    border-radius: var(--radius);
}

.search-card .form-control,
.search-card .input-group-text {
    border-color: var(--border);
    font-size: 0.95rem;
```

```
}

.search-card .form-control:focus {
    border-color: var(--primary);
    box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.1);
}
.contacts-table {
    font-size: 0.95rem;
}

.contacts-table thead th {
    background: var(--primary-light);
    color: var(--primary-dark);
    font-family: 'Poppins', sans-serif;
    font-weight: 600;
    font-size: 0.85rem;
    text-transform: uppercase;
    letter-spacing: 0.5px;
    padding: 0.85rem 1rem;
    border-bottom: 2px solid var(--primary);
}

.contacts-table tbody td {
    padding: 0.85rem 1rem;
    vertical-align: middle;
```

```
    border-color: var(--border);
}

.contacts-table tbody tr:hover {
    background-color: #f8faff;
}

/* Avatar circle with first letter */
.avatar-circle {
    width: var(--avatar-size);
    height: var(--avatar-size);
    border-radius: 50%;
    background: var(--avatar-bg);
    color: white;
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight: 700;
    font-size: 1rem;
    flex-shrink: 0;
    font-family: 'Poppins', sans-serif;
}

.contact-name-link {
    color: var(--text);
```

```
    text-decoration: none;
    font-weight: 600;
    transition: color 0.2s;
}

.contact-name-link:hover {
    color: var(--primary);
}

/* Action buttons */
.action-btns .btn {
    border-radius: 7px;
    padding: 0.3rem 0.55rem;
    font-size: 0.85rem;
    transition: all 0.15s;
}

.action-btns .btn:hover {
    transform: translateY(-1px);
    box-shadow: 0 2px 6px rgba(0,0,0,0.15);
}
.empty-state {
    padding: 4rem 2rem;
}
```

```
.empty-icon {
    font-size: 4rem;
}
.form-card {
    border: none;
    border-radius: var(--radius);
    overflow: hidden;
}

.form-card-header {
    background: linear-gradient(135deg, var(--primary) 0%, var(--primary-
dark) 100%);
    color: white;
    padding: 1.5rem;
}

.edit-header {
    background: linear-gradient(135deg, var(--warning) 0%, #d97706
100%);
}

.form-card .form-control {
    border-radius: 8px;
    border-color: var(--border);
    padding: 0.6rem 0.85rem;
```

```
    font-size: 0.95rem;
    transition: border-color 0.2s, box-shadow 0.2s;
}

.form-card .form-control:focus {
    border-color: var(--primary);
    box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.12);
}

.form-card .form-label {
    color: var(--text);
    margin-bottom: 0.4rem;
}

/* Buttons */
.btn-primary {
    background: var(--primary) !important;
    border-color: var(--primary) !important;
    border-radius: 8px;
    font-weight: 600;
    transition: all 0.2s;
}

.btn-primary:hover {
    background: var(--primary-dark) !important;
```

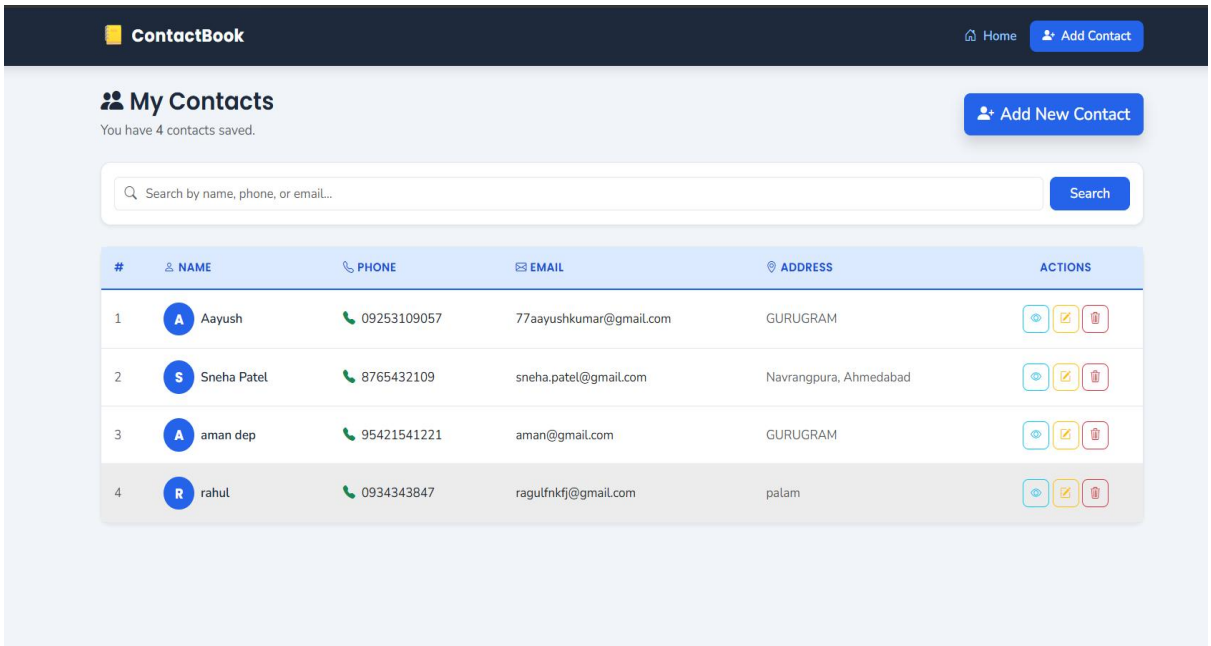
```
    transform: translateY(-1px);
    box-shadow: 0 4px 12px rgba(37, 99, 235, 0.3);
}

.btn-warning {
    border-radius: 8px;
    font-weight: 600;
}

.btn-outline-secondary {
    border-radius: 8px;
    font-weight: 600;
}
.detail-card {
    border: none;
    border-radius: var(--radius);
    overflow: hidden;
}

.detail-header {
    background: linear-gradient(135deg, #1e293b 0%, #334155 100%);
    color: white;
}

.detail-avatar {
```

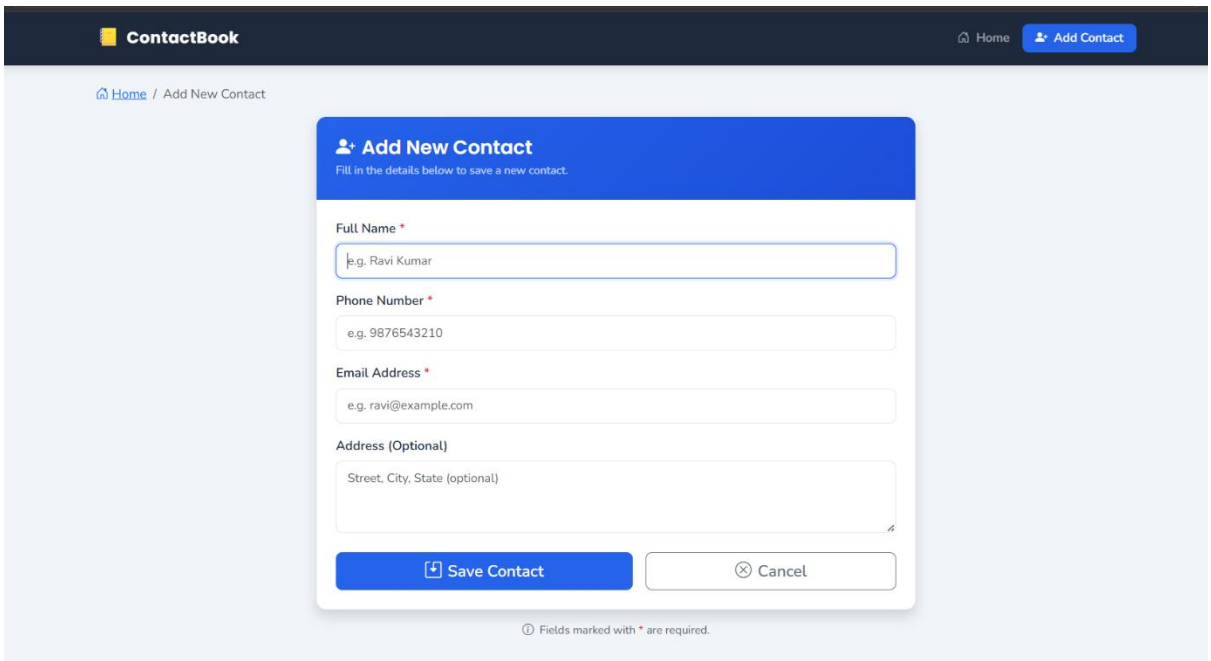
```
    width: 80px;
    height: 80px;
    border-radius: 50%;
    background: var(--primary);
    color: white;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 2rem;
    font-weight: 700;
    font-family: 'Poppins', sans-serif;
    border: 4px solid rgba(255,255,255,0.3);
}

.detail-list {
    display: flex;
    flex-direction: column;
    gap: 1rem;
}

.detail-item {
    display: flex;
    align-items: flex-start;
    gap: 1rem;
    padding: 0.85rem 1rem;
```

```
    background: var(--bg);
    border-radius: 10px;
    border: 1px solid var(--border);
}

.detail-icon {
    font-size: 1.3rem;
    margin-top: 2px;
    flex-shrink: 0;
}

.detail-label {
    font-size: 0.75rem;
    text-transform: uppercase;
    letter-spacing: 0.5px;
    color: var(--text-muted);
    font-weight: 600;
    margin-bottom: 2px;
}
.detail-value {
    font-weight: 600;
    color: var(--text);
    font-size: 1rem;
}
.contact-preview-box {
```

```
    background: #fff5f5;
    border: 1px solid #fed7d7;
}
.footer {
    background: var(--navbar-bg);
    color: #94a3b8;
    margin-top: auto;
}
.footer-icon {
    font-size: 1.1rem;
    margin-right: 4px;
}
.card {
    border-color: var(--border);
    border-radius: var(--radius);
}
@media (max-width: 576px) {
    .contacts-table thead th:nth-child(5),
    .contacts-table tbody td:nth-child(5) {
        display: none; /* Hide address column on small screens */
    }
    .page-title {
        font-size: 1.4rem;
    }
}
```

❖ OUTPUT :


ContactBook
Home
Add Contact
My Contacts
You have 4 contacts saved.
Add New Contact
Search by name, phone, or email...
Search
NAME PHONE EMAIL ADDRESS ACTIONS
1 A Aayush 09253109057 77aayushkumar@gmail.com GURUGRAM
2 S Sneha Patel 8765432109 sneha.patel@gmail.com Navrangpura, Ahmedabad
3 A aman dep 95421541221 aman@gmail.com GURUGRAM
4 R rahul 0934343847 ragulfnkfj@gmail.com palam


ContactBook
Home
Add Contact
Home / Add New Contact
Add New Contact
Fill in the details below to save a new contact.
Full Name *
e.g. Ravi Kumar
Phone Number *
e.g. 9876543210
Email Address *
e.g. ravi@example.com
Address (Optional)
Street, City, State (optional)
Save Contact
Cancel
Fields marked with * are required.

Experiment : 7

❖ Aim :

To design and develop an E-commerce Website using HTML, CSS, and JavaScript that allows users to view products, add items to cart, and simulate checkout functionality.

❖ Tasks :

• Project Setup & Introduction – Create a project folder named ecommerce_website and add HTML, CSS, and JavaScript files with proper linking and basic homepage.

• Product Listing Page – Display products with name, price, description, and Add to Cart button using cards or grid layout.

• Styling & Layout – Apply CSS for product cards, buttons, typography, and proper page layout.

• Shopping Cart Functionality – Add products to cart, display items dynamically, show quantity and total price, and allow item removal.

• Cart Summary & Price Calculation – Calculate individual item totals and overall cart total dynamically.

• Checkout Simulation – Create a checkout section to collect basic user details and display order confirmation message.

• Enhanced Features (Bonus) – Add quantity buttons, store cart data using browser storage, and improve UI with effects.

❖ Tech Stack Used :

• HTML • CSS • JavaScript

❖ Html Code :

>Index.html :

```
<section class="hero">
  <div class="hero-content">
    <h1 class="hero-title">Welcome to <span class="highlight">ShopEasy!</span> 🛍️ </h1>
    <p class="hero-subtitle">Discover amazing products at unbeatable prices. Quality you can trust!</p>
    <a href="#products" class="btn btn-hero">Shop Now ↓</a>
  </div>

  <!-- Fun decorative badges -->
  <div class="hero-badges">
    <div class="badge">🚚 Free Delivery</div>
    <div class="badge">🔒 Secure Payment</div>
    <div class="badge">⭐ Top Quality</div>
  </div>
</section>

<!-- ===== PRODUCTS SECTION ===== -->
<section class="products-section" id="products">

  <div class="section-header">
    <h2 class="section-title">🛒 Our Products</h2>
    <p class="section-subtitle">{{ products.count }} amazing products available!</p>
  </div>

  <!-- Products Grid -->
  <div class="products-grid">
```

```
{% for product in products %}
<!-- Product Card -->
<div class="product-card" id="product-{{ product.id }}">

    <!-- Product Emoji / Image Area -->
    <div class="product-image">
        <span class="product-emoji">{{ product.emoji }}</span>
        <!-- Category Badge -->
        <span class="category-badge">{{ product.get_category_display }}</span>
    </div>

    <!-- Product Info -->
    <div class="product-info">
        <h3 class="product-name">{{ product.name }}</h3>
        <p class="product-description">{{ product.description }}</p>

        <!-- Price and Button Row -->
        <div class="product-footer">
            <span class="product-price">₹{{ product.price }}</span>

            <!-- Add to Cart Button - calls addToCart() in script.js -->
            <button
                class="btn btn-add-cart"
                onclick="addToCart({{ product.id }}, '{{ product.name }}')"
                id="btn-{{ product.id }}"
            >
                + Add to Cart
            </button>
        </div>
```

```
        </div>

      </div>
      {% empty %}
      <!-- Shown if no products exist -->
      <div class="empty-message">
        <p>😟 No products available right now. Check back soon!</p>
      </div>
      {% endfor %}

    </div>
</section>

{% endblock %}
```

>base.html :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{% block title %}ShopEasy{% endblock %} | ShopEasy</title>

    {% load static %}

    <!-- Google Fonts -->
    <link rel="preconnect" href="https://fonts.googleapis.com">
```

```
    <link
href="https://fonts.googleapis.com/css2?family=Baloo+2:wght@400;500;600;700;800
&family=Nunito:wght@400;500;600;700&display=swap" rel="stylesheet">


    <!-- Our CSS file -->
    <link rel="stylesheet" href="{% static 'store/css/style.css' %}">
</head>
<body>


    <!-- ===== NAVBAR ===== -->
    <nav class="navbar">
        <div class="nav-container">


            <!-- Logo -->
            <a href="{% url 'index' %}" class="nav-logo">
                🛍️ ShopEasy
            </a>


            <!-- Navigation Links -->
            <div class="nav-links">
                <a href="{% url 'index' %}" class="nav-link">🏠 Home</a>
                <a href="{% url 'cart' %}" class="nav-link cart-link">
                    🛒 Cart
                    <!-- Cart count badge - shown only when cart has items -->
                    <span class="cart-badge" id="cart-badge">{{ cart_count }}</span>
                </a>
            </div>
        </div>
```

```
</nav>

<!-- ===== FLASH MESSAGES (Django messages framework) ===== -->
{% if messages %}
<div class="messages-container">
    {% for message in messages %}
    <div class="alert alert-{{ message.tags }}">
        {{ message }}
    </div>
    {% endfor %}
</div>
{% endif %}

<!-- ===== MAIN CONTENT (each page fills this) ===== -->
<main class="main-content">
    {% block content %}
    {% endblock %}
</main>

<!-- ===== FOOTER ===== -->
<footer class="footer">
    <div class="footer-container">
        <p>🛍️ <strong>ShopEasy</strong> — Your Friendly Online Store</p>
        <p>Made with ❤️ using Django | Experiment-7 Project</p>
    </div>
</footer>

<!-- Toast notification (for add-to-cart messages) -->
```

```
    <div class="toast" id="toast-message"></div>

    <!-- Our JavaScript file -->
    <script src="{% static 'store/js/script.js' %}"></script>

    <!-- Extra scripts from child templates -->
    {% block extra_js %}
    {% endblock %}

</body>
</html>
```

>cart.html :

```
<section class="cart-section">
    <div class="section-header">
        <h2 class="section-title">🛒 My Shopping Cart</h2>

        {% if cart_items %}
        <p class="section-subtitle">{{ cart_count }} item(s) in your cart</p>
        {% endif %}
    </div>

    {% if cart_items %}
```

```
<div class="cart-layout">

    <!-- ===== CART ITEMS TABLE ===== -->
    <div class="cart-items">

        {% for item in cart_items %}
        <!-- Cart Item Row -->
        <div class="cart-item" id="cart-item-{{ item.id }}">

            <!-- Product Emoji -->
            <div class="cart-item-emoji">{{ item.emoji }}</div>

            <!-- Product Name and Price -->
            <div class="cart-item-info">
                <h4 class="cart-item-name">{{ item.name }}</h4>
                <p class="cart-item-price">₹{{ item.price }} each</p>
            </div>

            <!-- Quantity Controls (+ / -) -->
            <div class="quantity-controls">
                <button
                    class="qty-btn qty-decrease"
                    onclick="updateCart('{{ item.id }}', 'decrease')"
                    title="Decrease quantity"
                >−</button>
```

```
                <span class="qty-display" id="qty-
{{ item.id }}">{{ item.quantity }}</span>

                <button
                    class="qty-btn qty-increase"
                    onclick="updateCart('{{ item.id }}', 'increase')"
                    title="Increase quantity"
                >+</button>
            </div>

            <!-- Item Total -->
            <div class="cart-item-total" id="item-total-{{ item.id }}">
                ₹{{ item.total }}
            </div>

            <!-- Remove Button -->
            <button
                class="btn-remove"
                onclick="removeFromCart('{{ item.id }}')"
                title="Remove item"
            >✕</button>

        </div>
        {% endfor %}
```

```
</div>

<!-- ===== CART SUMMARY BOX ===== -->
<div class="cart-summary">
    <h3 class="summary-title">📋 Order Summary</h3>

    <div class="summary-row">
        <span>Items ({{ cart_count }}):</span>
        <span id="summary-total">₹{{ cart_total }}</span>
    </div>

    <div class="summary-row">
        <span>Delivery:</span>
        <span class="free-tag">FREE ✅</span>
    </div>

    <div class="summary-divider"></div>

    <div class="summary-row total-row">
        <strong>Total:</strong>
        <strong id="grand-total">₹{{ cart_total }}</strong>
    </div>

    <!-- Proceed to Checkout -->
```

```
            <a href="{% url 'checkout' %}" class="btn btn-checkout">
                Proceed to Checkout →
            </a>

            <!-- Continue Shopping -->
            <a href="{% url 'index' %}" class="btn btn-outline">
                ← Continue Shopping
            </a>
        </div>

    </div>

    {% else %}

    <!-- Empty Cart Message -->
    <div class="empty-cart">
        <div class="empty-cart-icon">🛒</div>
        <h3>Your cart is empty!</h3>
        <p>Looks like you haven't added anything yet. Go explore our products!</p>
        <a href="{% url 'index' %}" class="btn btn-primary">Start Shopping →</a>
    </div>

    {% endif %}
```

```
</section>

{% endblock %}

{% block extra_js %}
<script>
    // CSRF token needed for Django POST requests
    const CSRF_TOKEN = '{{ csrf_token }}';
</script>
{% endblock %}
```

>checkout.html :

```
<section class="checkout-section">
    <div class="section-header">
        <h2 class="section-title">🛍️ Checkout</h2>
        <p class="section-subtitle">Almost there! Fill in your details to
complete the order.</p>
    </div>

    <div class="checkout-layout">

        <!-- ===== CHECKOUT FORM ===== -->
```

```
<div class="checkout-form-box">
    <h3 class="form-title">✋ Your Details</h3>

    <!--
        Form submits to place_order view.
        method="post" is required for Django forms.
        {% csrf_token %} is required by Django for security!
    -->
    <form action="{% url 'place_order' %}" method="post"
class="checkout-form" id="checkout-form">
        {% csrf_token %}

        <!-- Full Name -->
        <div class="form-group">
            <label for="name">👤 Full Name *</label>
            <input
                type="text"
                id="name"
                name="name"
                placeholder="e.g. Rahul Sharma"
                required
                class="form-input"
            >
        </div>
```

```
<!-- Email -->
<div class="form-group">
    <label for="email">📧 Email Address</label>
    <input
        type="email"
        id="email"
        name="email"
        placeholder="e.g. rahul@example.com"
        class="form-input"
    >
</div>

<!-- Address -->
<div class="form-group">
    <label for="address">🏠 Delivery Address *</label>
    <textarea
        id="address"
        name="address"
        placeholder="e.g. 123, MG Road, Koramangala"
        required
        class="form-input"
        rows="3"
    ></textarea>
</div>
```

```
<!-- City -->
<div class="form-group">
  <label for="city">🏙 City *</label>
  <input
    type="text"
    id="city"
    name="city"
    placeholder="e.g. Bengaluru"
    required
    class="form-input"
  >
</div>

<!-- Payment Note (Simulated) -->
<div class="payment-note">
  💳 <strong>Payment:</strong> This is a simulated checkout. No real payment required!
</div>

<!-- Submit Button -->
<button type="submit" class="btn btn-place-order">
  🎉 Place Order
</button>

</form>
```

```
    </div>
    <div class="checkout-summary">
        <h3 class="summary-title">🧾 Your Order</h3>

        {% for item in cart_items %}
        <div class="checkout-item">
            <span>{{ item.emoji }} {{ item.name }} ×
{{ item.quantity }}</span>
            <span>₹{{ item.total }}</span>
        </div>
        {% endfor %}

        <div class="summary-divider"></div>

        <div class="summary-row">
            <span>Delivery:</span>
            <span class="free-tag">FREE ✅</span>
        </div>

        <div class="summary-row total-row">
            <strong>Total Payable:</strong>
            <strong class="checkout-total">₹{{ cart_total }}</strong>
        </div>

        <!-- Back to Cart -->
```

```
            <a href="{% url 'cart' %}" class="btn btn-outline">← Edit
Cart</a>
        </div>

    </div>
</section>

{% endblock %}

{% block extra_js %}
<script>
    // Simple client-side form validation
    document.getElementById('checkout-form').addEventListener('submit',
function(e) {
        const name = document.getElementById('name').value.trim();
        const address = document.getElementById('address').value.trim();
        const city = document.getElementById('city').value.trim();

        if (!name || !address || !city) {
            e.preventDefault();
            alert('⚠ Please fill in all required fields!');
        }
    });
</script>
{% endblock %}
```

>conformation.html :

```
<section class="confirmation-section">

    <!-- Success Animation Box -->
    <div class="confirmation-box">

        <!-- Big success checkmark -->
        <div class="success-icon">✅</div>

        <h2 class="confirmation-title">Order Placed Successfully!</h2>
        <p class="confirmation-subtitle">
            Thank you, <strong>{{ name }}</strong>! 🎉 Your order is confirmed.
        </p>

        <!-- Order Number -->
        <div class="order-number-box">
            <p>Order Number</p>
            <h3 class="order-number">#{{ order_number }}</h3>
        </div>

        <!-- Delivery Info -->
        <div class="delivery-info">
```

```
        <p>🎁 Your order will be delivered to:</p>
        <p><strong>{{ address }}, {{ city }}</strong></p>
        {% if email %}
        <p>📧 Confirmation will be sent to:
<strong>{{ email }}</strong></p>
        {% endif %}
    </div>

    <!-- Order Items Summary -->
    <div class="order-summary-box">
        <h4> 📋 Order Summary</h4>

        {% for item in order_items %}
        <div class="order-item-row">
            <span>{{ item.emoji }} {{ item.name }} ×
{{ item.quantity }}</span>
            <span>₹{{ item.total }}</span>
        </div>
        {% endfor %}

        <div class="summary-divider"></div>

        <div class="order-item-row total-row">
            <strong>Total Paid:</strong>
            <strong class="confirm-total">₹{{ order_total }}</strong>
```

```
            </div>
        </div>

        <!-- Note about simulated payment -->
        <div class="simulation-note">
            ℹ This is a simulated order for project demonstration. No real payment was processed.
        </div>

        <!-- Action Buttons -->
        <div class="confirmation-actions">
            <a href="{% url 'index' %}" class="btn btn-primary">
                🛍 Continue Shopping
            </a>
        </div>

    </div>
</section>

{% endblock %}
```

>script.js:

```
    if (typeof CSRF_TOKEN !== 'undefined') {
        return CSRF_TOKEN;
    }
    // Fallback: read from cookie
    const name = 'csrftoken';
    const cookies = document.cookie.split(';');
    for (let cookie of cookies) {
        const [key, value] = cookie.trim().split('=');
        if (key === name) return decodeURIComponent(value);
    }
    return '';
}

function showToast(message, type = 'success') {
    const toast = document.getElementById('toast-message');
    if (!toast) return;

    // Set message text and style
    toast.textContent = message;
    toast.className = `toast show ${type}`;

    // Auto-hide after 3 seconds
    setTimeout(() => {
        toast.classList.remove('show');
    }, 3000);
}
function updateCartBadge(count) {
    const badge = document.getElementById('cart-badge');
```

```
    if (badge) {
        badge.textContent = count;

        // Hide badge if cart is empty
        if (count <= 0) {
            badge.style.display = 'none';
        } else {
            badge.style.display = 'flex';
        }
    }
}
function addToCart(productId, productName) {
    const button = document.getElementById(`btn-${productId}`);
    if (button) {
        button.textContent = 'Adding...';
        button.disabled = true;
    }
    fetch('/add-to-cart/', {
        method: 'POST',
        headers: {
            'Content-Type': 'application/json',
            'X-CSRFToken': getCsrfToken(),  // Required by Django!
        },
        body: JSON.stringify({ product_id: productId }),
    })
    .then(response => response.json())  // Parse JSON response
    .then(data => {
        if (data.success) {
            // Show success toast message
```

```
        showToast(`☑ ${productName} added to cart!`, 'success');
        updateCartBadge(data.cart_count);
        if (button) {
            button.textContent = 'Added ✓';
            button.classList.add('added');

            // Reset button after 2 seconds
            setTimeout(() => {
                button.textContent = '+ Add to Cart';
                button.classList.remove('added');
                button.disabled = false;
            }, 2000);
        }
    } else {
        // Show error message
        showToast(`❌ ${data.message}`, 'error');
        if (button) {
            button.textContent = '+ Add to Cart';
            button.disabled = false;
        }
    }
})
.catch(error => {
    // Network or other error
    console.error('Add to cart error:', error);
    showToast('❌ Something went wrong. Please try again.', 'error');
    if (button) {
        button.textContent = '+ Add to Cart';
        button.disabled = false;
```

```
        }
    });
}
function removeFromCart(productId) {

    if (!confirm('Remove this item from your cart?')) return;

    fetch('/remove-from-cart/', {
        method: 'POST',
        headers: {
            'Content-Type': 'application/json',
            'X-CSRFToken': getCsrfToken(),
        },
        body: JSON.stringify({ product_id: productId }),
    })
    .then(response => response.json())
    .then(data => {
        if (data.success) {
            // Remove the item row from the DOM (no page reload!)
            const itemElement = document.getElementById(`cart-item-${productId}`);
            if (itemElement) {
                // Fade out animation before removing
                itemElement.style.transition = 'opacity 0.3s, transform 0.3s';
                itemElement.style.opacity = '0';
                itemElement.style.transform = 'translateX(-20px)';
                setTimeout(() => itemElement.remove(), 300);
            }

            updateCartTotal(data.cart_total, data.cart_count);
```

```
            updateCartBadge(data.cart_count);

            showToast('🗑️ Item removed from cart', 'success');

            // If cart is now empty, reload to show empty cart message
            if (data.cart_count === 0) {
                setTimeout(() => location.reload(), 800);
            }
        } else {
            showToast(`❌ ${data.message}`, 'error');
        }
    })
    .catch(error => {
        console.error('Remove from cart error:', error);
        showToast('❌ Something went wrong.', 'error');
    });
}
function updateCart(productId, action) {
    fetch('/update-cart/', {
        method: 'POST',
        headers: {
            'Content-Type': 'application/json',
            'X-CSRFToken': getCsrfToken(),
        },
        body: JSON.stringify({ product_id: productId, action: action }),
    })
    .then(response => response.json())
    .then(data => {
```

```
if (data.success) {
    if (data.removed) {
        // Item quantity reached 0 - remove the row
        const itemElement = document.getElementById(`cart-item-${productId}`);
        if (itemElement) {
            itemElement.style.transition = 'opacity 0.3s';
            itemElement.style.opacity = '0';
            setTimeout(() => itemElement.remove(), 300);
        }
        showToast('🗑 Item removed from cart', 'success');

        // Reload if cart is empty
        if (data.cart_count === 0) {
            setTimeout(() => location.reload(), 800);
        }
    } else {
        // Update quantity display
        const qtyDisplay = document.getElementById(`qty-${productId}`);
        if (qtyDisplay) {
            qtyDisplay.textContent = data.quantity;
            // Small bounce animation on quantity change
            qtyDisplay.style.transform = 'scale(1.4)';
            setTimeout(() => { qtyDisplay.style.transform = 'scale(1)'; }, 200);
        }

        // Update item total
        const itemTotal = document.getElementById(`item-total-${productId}`);
        if (itemTotal) {
            itemTotal.textContent = `₹${data.item_total}`;
```

```
            }
        }

        // Update the cart summary totals
        updateCartTotal(data.cart_total, data.cart_count);

        // Update navbar badge
        updateCartBadge(data.cart_count);
      }
    })
    .catch(error => {
      console.error('Update cart error:', error);
      showToast('❌ Something went wrong.', 'error');
    });
}
function updateCartTotal(newTotal, itemCount) {
    // Update summary box total
    const summaryTotal = document.getElementById('summary-total');
    if (summaryTotal) summaryTotal.textContent = `₹${newTotal}`;

    // Update grand total
    const grandTotal = document.getElementById('grand-total');
    if (grandTotal) grandTotal.textContent = `₹${newTotal}`;
    const subtitle = document.querySelector('.section-subtitle');
    if (subtitle && itemCount !== undefined) {
      subtitle.textContent = `${itemCount} item(s) in your cart`;
    }
}
document.addEventListener('DOMContentLoaded', function () {
```

```
    // Hide badge if it shows "0"
    const badge = document.getElementById('cart-badge');
    if (badge && (badge.textContent === '0' || badge.textContent === '')) {
        badge.style.display = 'none';
    }
    const qtyDisplays = document.querySelectorAll('.qty-display');
    qtyDisplays.forEach(el => {
        el.style.transition = 'transform 0.2s ease';
    });
    console.log('☑ ShopEasy script.js loaded successfully!');
});
```

❖ CSS Code :

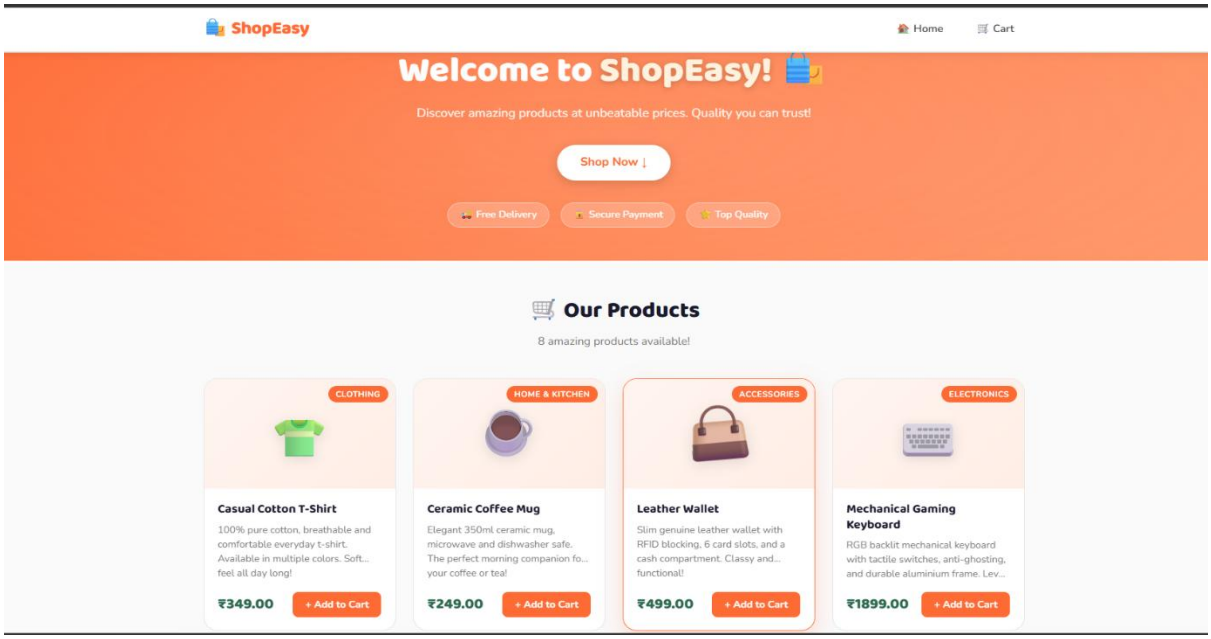
```
/* Reset */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
/* Body */
body {
  font-family: Arial, sans-serif;
  background: #f5f5f5;
}
/* Navbar */
.navbar {
  background: orange;
  color: white;
  padding: 15px;
```

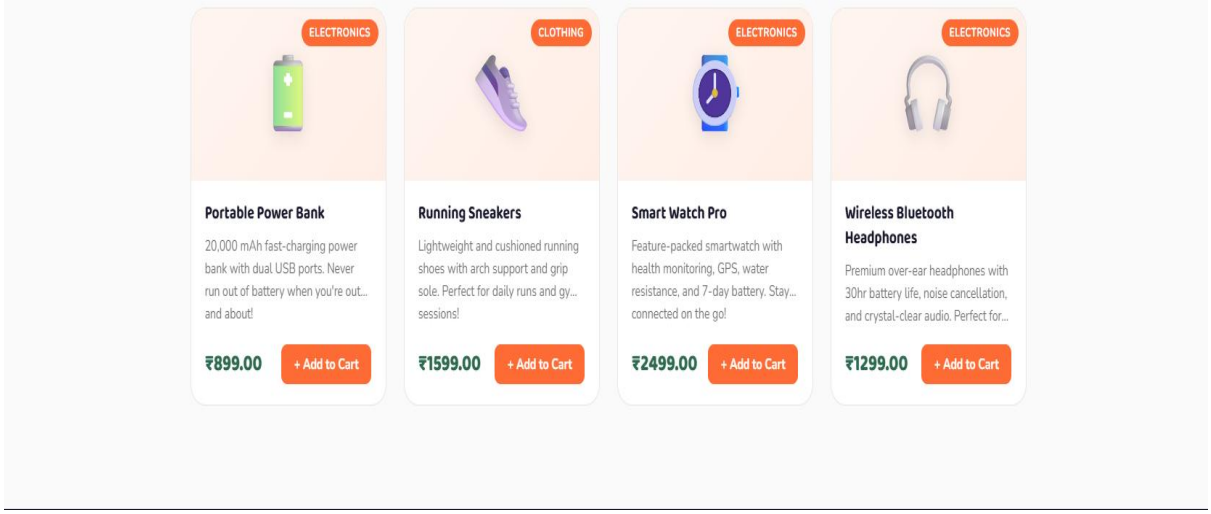
```
  display: flex;
  justify-content: space-between;
}
.navbar a {
  color: white;
  margin-left: 10px;
  text-decoration: none;
}

/* Hero Section */
.hero {
  background: orange;
  color: white;
  text-align: center;
  padding: 40px;
}
.hero button {
  padding: 10px 20px;
  background: white;
  color: orange;
  border: none;
  margin-top: 10px;
}
/* Products */
.products {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 15px;
  padding: 20px;
```

```
}
.product {
  background: white;
  padding: 15px;
  text-align: center;
  border: 1px solid #ddd;
}
.product button {
  margin-top: 10px;
  padding: 8px;
  background: orange;
  color: white;
  border: none;
}
/* Cart */
.cart {
  padding: 20px;
}
.cart-item {
  display: flex;
  justify-content: space-between;
  background: white;
  padding: 10px;
  margin-bottom: 10px;
}
/* Checkout */
.checkout {
  padding: 20px;
}
```

```
input {
  width: 100%;
  padding: 8px;
  margin: 5px 0;
}
/* Footer */
.footer {
  background: black;
  color: white;
  text-align: center;
  padding: 10px;
}
/* Mobile */
@media (max-width: 768px) {
  .products {
    grid-template-columns: 1fr;
  }
}
```

❖OUTPUT:


ShopEasy
Home
Cart
Welcome to ShopEasy!
Discover amazing products at unbeatable prices. Quality you can trust!
Shop Now ↓
Free Delivery
Secure Payment
Top Quality
Our Products
8 amazing products available!
CLOTHING
Casual Cotton T-Shirt
100% pure cotton, breathable and comfortable everyday t-shirt. Available in multiple colors. Soft... feel all day long!
₹349.00
+ Add to Cart
HOME & KITCHEN
Ceramic Coffee Mug
Elegant 350ml ceramic mug, microwave and dishwasher safe. The perfect morning companion fo... your coffee or tea!
₹249.00
+ Add to Cart
ACCESSORIES
Leather Wallet
Slim genuine leather wallet with RFID blocking, 6 card slots, and a cash compartment. Classy and... functional!
₹499.00
+ Add to Cart
ELECTRONICS
Mechanical Gaming Keyboard
RGB backlit mechanical keyboard with tactile switches, anti-ghosting, and durable aluminium frame. Lev...
₹1899.00
+ Add to Cart


ELECTRONICS
Portable Power Bank
20,000 mAh fast-charging power bank with dual USB ports. Never run out of battery when you're out... and about!
₹899.00
+ Add to Cart
CLOTHING
Running Sneakers
Lightweight and cushioned running shoes with arch support and grip sole. Perfect for daily runs and gy... sessions!
₹1599.00
+ Add to Cart
ELECTRONICS
Smart Watch Pro
Feature-packed smartwatch with health monitoring, GPS, water resistance, and 7-day battery. Stay... connected on the go!
₹2499.00
+ Add to Cart
ELECTRONICS
Wireless Bluetooth Headphones
Premium over-ear headphones with 30hr battery life, noise cancellation, and crystal-clear audio. Perfect for...
₹1299.00
+ Add to Cart